

From one cell to a population: burst-aware renewal dynamics and the bursting–budding comparison

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1 What we need from Chapters 3 and 4a

This chapter embeds the single-cell theory of Chapters 3 and 4a in a population. Nothing new from the single cell is required — but several of its closed forms will be quoted constantly, and they are collected here once, boxed, with the labels the rest of the chapter uses. Chapter 3 supplies the derivations; Chapter 4a the distribution theory; this section is the reference card.

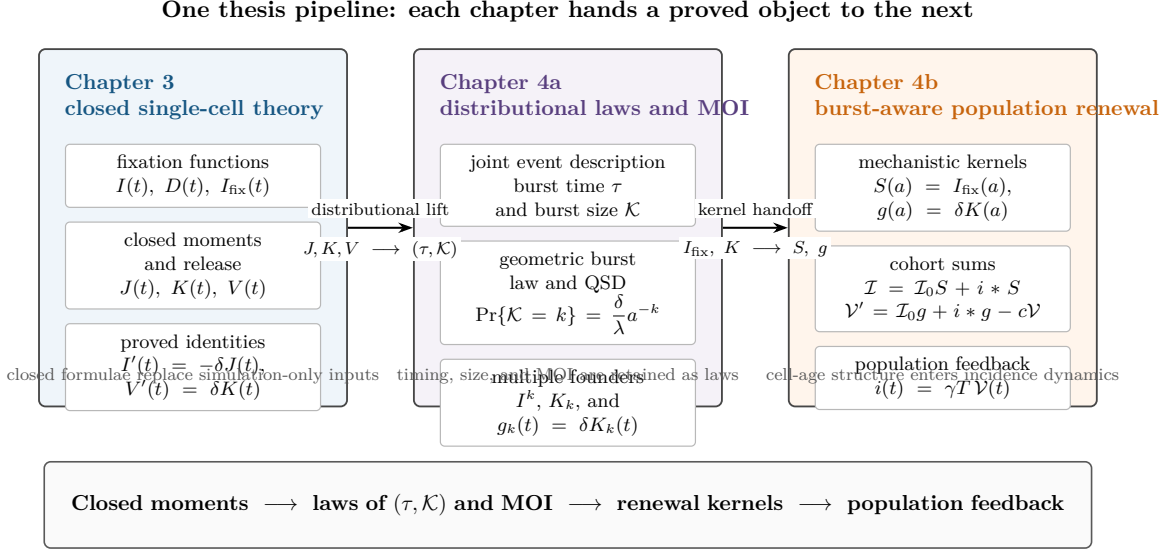


Figure 1: Logical handoff across the trilogy: closed single-cell moments and fixation functions (Chapter 3), distributional laws and multiplicity (Chapter 4a), and burst-aware renewal kernels in population dynamics (Chapter 4b).

Throughout: λ, μ, δ are the intracellular birth, death and catastrophe rates per replicative unit; H is the absorbing rupture state; $\mathbf{X}_t$ the intracellular load, $\mathbf{W}_t$ the released count, τ the burst time, and $\mathcal{K}$ the burst size. We write $w = e^{\theta t}$, and one naming decision is made now and kept: the burst-time density, which is $\delta J(t)$ in both earlier chapters, is written as δJ in this chapter, because φ is reserved for the partial-release fraction of §6.5.

1.1 Roots and identities

The roots of the quadratic $\lambda f^2 - (\lambda + \mu + \delta)f + \mu$ are

$$a = \eta + \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad \eta = \frac{\lambda + \mu + \delta}{2\lambda}, \quad (1)$$

with $b < 1 < a$, and the derived constants are

$$A = a - 1, \quad B = 1 - b, \quad \theta = \lambda(a - b), \quad \kappa = 1 + \frac{\delta}{2\lambda}. \quad (2)$$

The identities that simplify everything downstream are

$$ab = \frac{\mu}{\lambda}, \quad a + b = \frac{\lambda + \mu + \delta}{\lambda}, \quad AB = \frac{\delta}{\lambda}, \quad A + B = a - b, \quad \theta = \lambda(a - b). \quad (3)$$

1.2 The three fixation functions

The no-burst probability, the internal-extinction probability, and the non-fixation probability are

$$\boxed{I(t) = \frac{aB + bAw}{B + Aw}, \quad D(t) = \frac{ab(w-1)}{aw-b}, \quad I_{\text{fix}}(t) = I(t) - D(t) = \frac{(a-b)^2 w}{(B + Aw)(aw-b)}. \quad (4)}$$

$I(\infty) = D(\infty) = b - b$ is the probability that the cell clears internally without ever bursting — while $I_{\text{fix}}(0) = 1$, $I'_{\text{fix}}(0) = -(\mu + \delta)$ and $I_{\text{fix}}(\infty) = 0$. The population survival kernel of §3 is I_{fix} itself.

1.3 Moments and mean release

With $I' = -\delta J$ and $V' = \delta K$,

$$\boxed{J(t) = \frac{(a-b)^2 w}{(B + Aw)^2} = -\frac{\lambda}{\delta}(I - a)(I - b), \quad (5)}$$

$$\boxed{K(t) = \left[1 + \frac{2\lambda}{\delta}(1 - I)\right]J = \frac{2\lambda}{\delta}(\kappa - I)J, \quad (6)}$$

and the cumulative mean release, a quadratic in $1 - I$ alone,

$$\boxed{V(t) = (1 - I)\left[1 + \frac{\lambda}{\delta}(1 - I)\right]. \quad (7)}$$

Its late-time value, the total expected output of one cell, is

$$\boxed{V_{\infty} = \frac{a(1-b)}{a-1} = \frac{aB}{A} = \frac{\lambda - \mu}{\delta}(1-b) + 1, \quad \mathbb{E}(\mathcal{K} \mid \text{burst}) = \frac{V_{\infty}}{1-b} = \frac{a}{a-1}. \quad (8)}$$

Note again that V_{∞} counts the internally cleared cells with their release of zero; the conditional mean removes them.

1.4 Distribution-level facts

Three results of Chapter 4a are used as black boxes here. First, the burst-time density is

$$-I'(t) = \delta J(t), \quad (9)$$

defective with total mass $1 - b$. Second, the burst size, unconditional, obeys

$$\Pr\{\mathcal{K} = k\} = \frac{\delta}{\lambda} a^{-k}, \quad k \geq 1, \quad (10)$$

with the mass b at “no burst”; conditioned on bursting this becomes $(a-1)a^{-k}$, which is exactly the quasi-stationary distribution of the load (both are geometric on $\{1, 2, \dots\}$ with ratio $1/a$). Third, given that the burst occurs at time t , its size is a size-biased draw from the state probabilities:

$$\boxed{\mathbb{E}(\mathcal{K} \mid \tau = t) = \frac{K(t)}{J(t)} = 1 + \frac{2\lambda}{\delta}(1 - I(t)), \quad (11)}$$

tending at late time to $(a+1)/(a-1)$, the size-biased quasi-stationary mean.

1.5 The second moment of the release

The whole-host variance prediction of §8.3 starts from

$$\mathbb{E}(\mathbf{W}_t^2) = \frac{2(\lambda - \mu)}{\delta} V - \frac{\lambda + \mu}{\delta} I - K + \frac{\lambda + \mu}{\delta} + 1, \quad (12)$$

and $\text{Var}(\mathbf{W}_t) = \mathbb{E}(\mathbf{W}_t^2) - V^2$.

1.6 The mean productive lifetime

The integral of the survival kernel, used in §4 to anchor the effective parameters, is

$$\mathbb{E}(T_{\text{prod}}) = \int_0^\infty I_{\text{fix}}(t) dt = \frac{1}{\lambda} \log \frac{a}{a-1} = \frac{1}{\lambda} \log \mathbb{E}(\mathcal{K} \mid \text{burst}). \quad (13)$$

1.7 Multiplicity of infection

A cell founded by k particles releases superlinearly more. Its moments and release flux are

$$J_k(t) = kJ(t)I(t)^{k-1}, \quad K_k(t) = kK(t)I(t)^{k-1} + k(k-1)J(t)^2I(t)^{k-2}, \quad (14)$$

$$g_k(t) = \delta K_k(t) = \delta \left(kK I^{k-1} + k(k-1)J^2 I^{k-2} \right), \quad (15)$$

and its lifetime yield, for $k \geq 2$,

$$V_\infty^{(k)} = (1 - b^k) + \frac{2\lambda}{\delta} \left[\frac{1}{k+1} - b^k + \frac{k b^{k+1}}{k+1} \right] + k(k-1) \frac{\lambda}{\delta} \int_1^b (x-a)(x-b) x^{k-2} dx, \quad (16)$$

simplifying to $V_\infty^{(k)} = k + \lambda/\delta$ when $\mu = 0$. The conditional mean burst $V_\infty^{(k)}/(1 - b^k)$ increases with k : multiply infected cells burst more often, and bigger.

Scope

Everything in this section is quoted; the derivations live in Chapters 3 and 4a, and the verification of the population-level consequences of what follows is recorded in Appendix B. The population notation of this chapter — $\mathcal{I}$, $\mathcal{V}$, γ , c , T , p , $d_{\mathcal{I}}$ — is chosen to avoid collision with all of the above, and is collected in Table 1.

Symbol	Meaning	Notes
T	target-cell count	held fixed
γ	infection rate	
$\mathcal{I}, \mathcal{V}$	infected cells, free particles	population counts
c	virion clearance	
$d_{\mathcal{I}}$	infected-cell removal (phenomenological)	not δ
$p, p_{\text{eff}}(r)$	constant / effective release rate	
$g(a)$	release kernel, $\delta K(a)$	expected flux at age a
$S(a)$	survival kernel, $I_{\text{fix}}(a)$	
q	$\gamma T / (\gamma T + c)$, infection success	branching
m	$q V_{\infty}$, branching mean offspring	
L	$a(1 - b) = a - \mu/\lambda$, flooding parameter	
φ	boolean partial-release fraction (§6.5)	
ζ	intensity decay rate of Model 10 (§6.4)	not κ

Table 1: Population-level notation of this chapter.

2 From one cell to a population: why a constant release rate fails

The single-cell theory is complete. We now ask what it implies at the level of a population of infected cells — and begin with the standard object it is meant to improve upon.

2.1 The classical BMVR as comparator

In the simplified (target-cell-constant) Basic Model of Viral Replication [1, 2, 3], infected cells $\mathcal{I}$ and free particles $\mathcal{V}$ obey

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad (17)$$

with target cells T held fixed, infection rate γ , virion clearance c , and where the two constants that concern us are p , the per-cell release rate, and $d_{\mathcal{I}}$, the infected-cell removal rate. Both are phenomenological: they are parameters to be fitted, not quantities derived from anything happening inside the cell. The basic reproduction number of (17) is

$$R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}. \quad (18)$$

Every cell, regardless of its age or its contents, releases at the same rate p and dies at the same rate $d_{\mathcal{I}}$. For a budding virus this is at least a plausible picture; for a lytic pathogen it is plainly false. It is worth making the budding picture explicit, because it completes the comparator that the rest of the chapter is built against.

A budding infected cell experiences two *independent* event types: release of one infectious particle, as a Poisson process of rate p , and death of the cell, as a Poisson process of rate $d_{\mathcal{I}}$. The two continue side by side until the cell dies. Its single-cell survival and cumulative release are then

$$I_{\text{bud}}(t) = e^{-d_{\mathcal{I}} t}, \quad V_{\text{bud}}(t) = \frac{p}{d_{\mathcal{I}}} \left(1 - e^{-d_{\mathcal{I}} t}\right), \quad \frac{dV_{\text{bud}}}{dI_{\text{bud}}} = -\frac{p}{d_{\mathcal{I}}} : \quad (19)$$

the mean cell lifetime is $1/d_I$, the lifetime release is p/d_I , and the total number a cell releases is geometric on $\{0, 1, 2, \dots\}$ with $\Pr\{\mathcal{K}_{\text{bud}} = k\} = \frac{d_I}{d_I + p} \left(\frac{p}{d_I + p}\right)^k$ — the number of successes before the terminating death in a sequence of events of which each is a release with probability $p/(d_I + p)$. All of this is contained in the two constants (p, d_I) , which is exactly why the classical model is a faithful one-cell limit for a budding pathogen.

The structural contrast with bursting is precise, and it is the engine of everything that follows. In budding, release and death are independent events: a cell may release many particles and then die with whatever it contains lost, or release nothing at all. In bursting, release and death are the *same* event: whatever has accumulated is released at once, and the cell's productive life ends in that instant. The single-cell comparison in detail — release laws, lifetimes, conditional means — is in Chapter 4a; here the point is that (17) encodes the budding picture exactly, and the question of this part of the chapter is what it should be replaced with when release comes in terminal batches.

2.2 The synchronous cohort

Consider first a clean special case: N_0 cells, all infected at the same instant, each running an independent copy of the single-cell process, with no reabsorption or extracellular replication. The number still productively infected at time t is $N_0 I_{\text{fix}}(t)$ (exactly in expectation, and with vanishing relative error as N_0 grows), and the expected cumulative extracellular count is simply

$$\mathcal{V}(t) = N_0 V(t), \quad (20)$$

since $V(t)$ is the expected release from one cell. In this synchronous picture, the release *rate* is $N_0 V'(t) = N_0 \delta K(t)$: it depends on the age of the cohort, and it is not constant. The classical model's pI can mimic such a cohort only by accident, and only briefly.

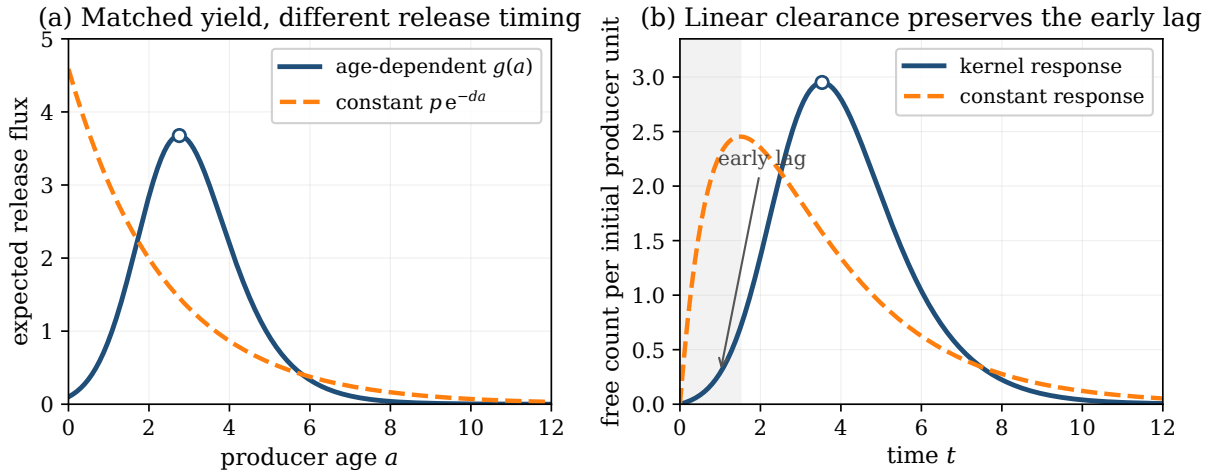


Figure 2: Age-dependent release compared with a constant-production surrogate for a synchronous cohort of producer units. Both schedules have $V_\infty = 11$ and $\mathbb{E}[T] = 2.398$, with $d = 1/\mathbb{E}[T]$, $p = V_\infty d$, $(\lambda, \mu, \delta) = (1, 0, 0.1)$, and clearance rate $c = 1$. (a) The release kernel $g(a) = \delta K(a)$ is delayed relative to $p e^{-da}$ despite the matched lifetime yield. (b) Linear clearance preserves the resulting early lag in the free-count response.

2.3 The proposal that cannot work

The natural thought — and the one originally pursued — is to replace the constant p in (17) by a constant built from the intracellular theory: the quasi-stationary mean,

$$p \stackrel{?}{=} \langle X \rangle_{\text{QS}} = \frac{a}{a-1}. \quad (21)$$

The intuition is that a long-lived infected cell hovers at the QS level, so its bursting should look like release at rate $\langle X \rangle_{\text{QS}}$. The obstruction is dimensional: p has units of particles per unit time, while $\langle X \rangle_{\text{QS}}$ is a pure count. Substituting (21) introduces a spurious factor of time into the model. Worse, when $\mu = 0$ one has $\langle X \rangle_{\text{QS}} = V_\infty$ exactly, so the proposal substitutes the *total lifetime output* of a cell in place of its *output rate*.

Two repairs suggest themselves, and both fail for related reasons. First, average the burst size over the burst-time density δJ , the density of τ derived in Chapter 4a:

$$\int_0^\infty \delta J(t) \mathbb{E}(K \mid \tau = t) dt = \delta \int_0^\infty K(t) dt = V_\infty, \quad (22)$$

using (11) and $V' = \delta K$ — again a count, not a rate. Second, weight the load-dependent flux by the burst-time density, $\delta \int \delta J(t) \mathbb{E}(\mathbf{X}_t^2 \mid \cdot) dt$: this has the right dimensions, but it averages over the wrong measure. The burst-time density describes *when a single cell bursts*; what a population release rate needs is an average over the *ages of the cells currently infected*. In an asynchronous infection those two distributions are not the same, and no reweighting of a single-cell count by a single-cell density can fix the substitution, because the difficulty is not in the single-cell theory at all. It is in the population bookkeeping.

2.4 What the attempt got right

Strip the failed proposal back and something correct remains. Suppose a single infected macrophage founds a chain of infections: when it ruptures, each released particle goes on to found a new intracellular process in a new cell; these in turn end at different times, and the cycles continue, beginning and ending, beginning and ending. After a sufficient time there emerges a kind of stationarity: any given release event exhibits a burst time and a burst size distributed according to the laws of Chapter 4a, and short-duration processes occur more frequently than long ones simply because they turn over faster. What we want is the overall rate of release events and their sizes — the quantity that should replace the $p\mathcal{I}$ term of (17).

Two observations from §2.2–§2.3 organise what follows. First, the expected release flux of a cell of age a is not a constant but $\delta K(a)$ — the burst rate $\delta \mathbf{X}_t$ times the size $\mathbf{X}_t$, averaged. Second, at any host time the infected population is a mixture of ages, one cohort for each past moment of infection. The correct population model must therefore keep track of *cell age*: how many cells were infected when, how many of them are still productive, and how much each age class is releasing. That is precisely an age-structured, or renewal, formulation, and because both of its ingredients already exist in closed form, it is exactly solvable. We take it up next.

3 Burst-aware renewal viral dynamics

3.1 The two kernels

Age-structured population dynamics needs exactly two single-cell ingredients: how long a cell stays productive, and how much it releases at each age. Both are already in closed form.

The *survival kernel* is the probability that a cell infected a time units ago is still productively infected:

$$S(a) := I_{\text{fix}}(a) = \frac{(a - b)^2 e^{\theta a}}{(B + Ae^{\theta a})(ae^{\theta a} - b)}. \quad (23)$$

The *release kernel* is the expected release flux at cell-age a :

$$g(a) := \delta K(a) = V'(a). \quad (24)$$

The reasoning behind (24) is the one that the original attempt stated and then failed to use (§2.3): a cell bursts at rate $\delta \mathbf{X}_t$ and, when it does, releases $\mathbf{X}_t$ particles, so the expected number of particles released in $[a, a + da]$ is $\delta \mathbb{E}(\mathbf{X}_t^2) da = \delta K(a) da$. With K from (6), both kernels are explicit functions of age — this is what it means for the population model to be *forced by* the intracellular process (Figure 3).

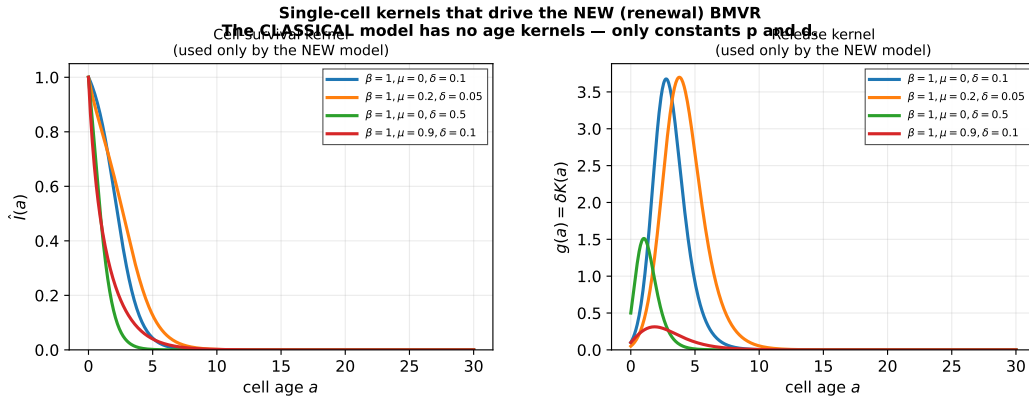


Figure 3: The cell-survival kernel $I_{\text{fix}}(a)$ and the release kernel $g(a) = \delta K(a)$ for four intracellular parameter sets. These age-dependent kernels drive the renewal model of this section; the classical model (17) has no kernels, only the constants p and $d_{\mathcal{I}}$.

The same intracellular kernels change character across the flooding boundary

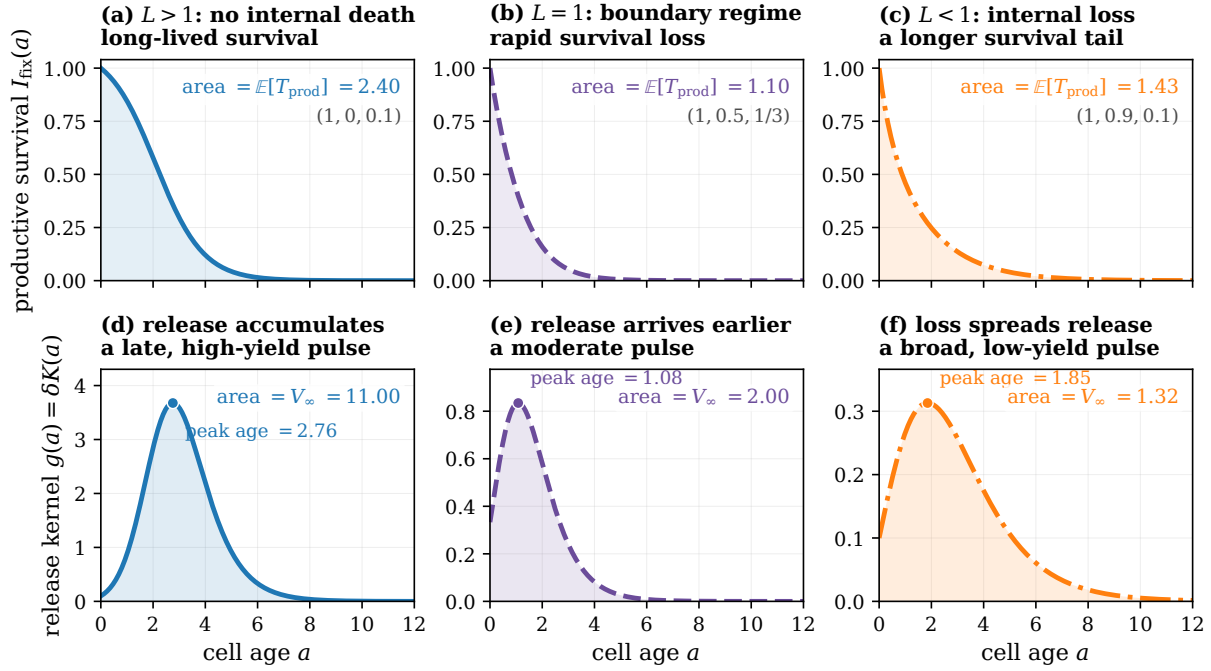


Figure 4: The exact survival and release kernels across the three flooding regimes used in §5. From left to right, $(\lambda, \mu, \delta) = (1, 0, 0.1)$, $(1, 0.5, 1/3)$, and $(1, 0.9, 0.1)$ give $L = 1.10$, 1.00 , and 0.42 . The shaded areas verify $\int_0^{\infty} I_{\text{fix}}(a) da = \mathbb{E}(T_{\text{prod}})$ and $\int_0^{\infty} g(a) da = V_{\infty}$ in each column. Each release panel has its own vertical scale so that the timing shift remains visible despite the large yield difference: the three values of V_{∞} are 11.00, 2.00, and 1.32. Population conclusions therefore inherit the full kernel shapes, not only their integrals.

3.2 The renewal system

Hold target cells T fixed, let infection occur at rate γ per virion, clear virions at rate c , and define the incidence

$$i(t) = \gamma T \mathcal{V}(t). \quad (25)$$

Each infection event at time $t - \alpha$ starts a new cell of age α ; such a cell is still productive with probability $I_{\text{fix}}(\alpha)$ and releases at expected rate $g(\alpha)$. Summing over all infection times, and adding the contribution $\mathcal{I}_0$ of cells already infected at $t = 0$,

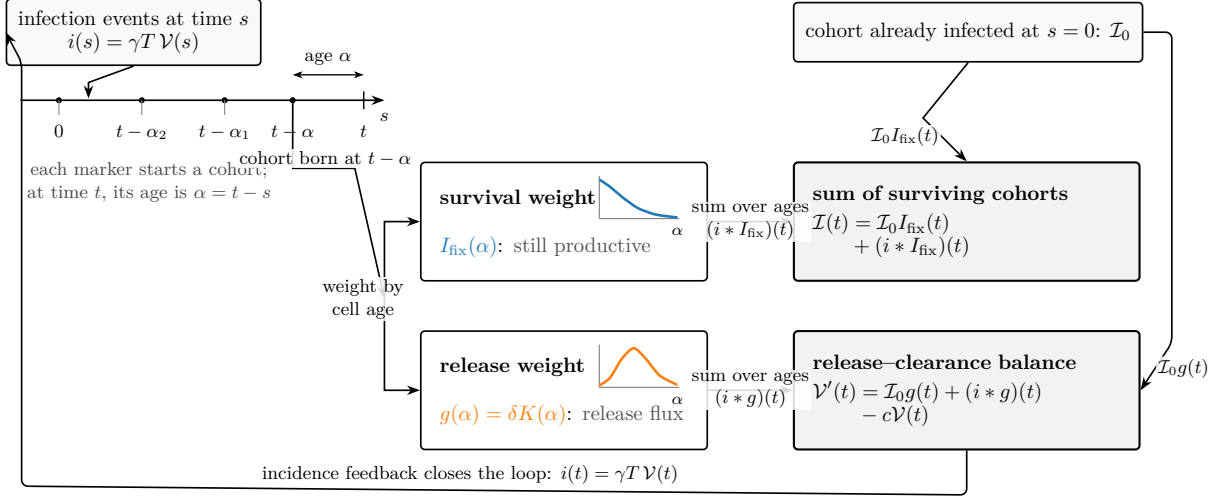


Figure 5: Renewal bookkeeping for burst-aware BMVR. Infection cohorts age from their infection times; weighting incidence by cell survival $I_{\text{fix}}(\alpha)$ and release $g(\alpha) = \delta K(\alpha)$, then summing over ages, gives $\mathcal{I}(t) = \mathcal{I}_0 I_{\text{fix}}(t) + (i * I_{\text{fix}})(t)$ and $\mathcal{V}'(t) = \mathcal{I}_0 g(t) + (i * g)(t) - c \mathcal{V}(t)$. The feedback $i(t) = \gamma T \mathcal{V}(t)$ closes the population loop.

Theorem 3.1 (New viral dynamics — burst-aware BMVR).

$$\mathcal{I}(t) = \mathcal{I}_0 I_{\text{fix}}(t) + \int_0^t i(t - \alpha) I_{\text{fix}}(\alpha) d\alpha = \mathcal{I}_0 I_{\text{fix}} + (i * I_{\text{fix}})(t), \quad (26)$$

$$\frac{d\mathcal{V}}{dt}(t) = \mathcal{I}_0 g(t) + \int_0^t i(t - \alpha) g(\alpha) d\alpha - c \mathcal{V}(t) = \mathcal{I}_0 g + (i * g)(t) - c \mathcal{V}(t). \quad (27)$$

The correspondence with the classical model is:

Classical BMVR (17)	New system (26)–(27)
$\mathcal{I}' = i - d_{\mathcal{I}} \mathcal{I}$ (constant removal rate $d_{\mathcal{I}}$)	$\mathcal{I} = \mathcal{I}_0 I_{\text{fix}} + i * I_{\text{fix}}$ (age-structured survival via I_{fix})
$\mathcal{V}' = p \mathcal{I} - c \mathcal{V}$ (constant release rate p)	$\mathcal{V}' = \mathcal{I}_0 g + i * g - c \mathcal{V}$ (age-structured release via $g = \delta K$)

Both constant terms are replaced, not just the release: the memoryless removal $d_{\mathcal{I}} \mathcal{I}$ becomes the age structure implicit in (26), and the constant release $p \mathcal{I}$ becomes the convolution with g . A replacement that touched only the release term — the natural first thought of §2.3 — would leave the removal side still memoryless, and still wrong.

Differentiating (26) makes the structure visible:

$$\frac{d\mathcal{I}}{dt} = i(t) + \mathcal{I}_0 I'_{\text{fix}}(t) + \int_0^t i(t - \alpha) I'_{\text{fix}}(\alpha) d\alpha, \quad (28)$$

in which the incidence appears as a source exactly as in the classical model, while the remaining terms encode age-dependent loss of productivity. There is, in general, no constant $d_{\mathcal{I}}$ for which this reduces to $\mathcal{I}' = i - d_{\mathcal{I}} \mathcal{I}$ for every history of i : such a reduction holds only when everything grows exponentially, and that exception is exactly what §4 exploits.

Remark 3.2 (What is and is not new). The renewal, or age-structured, framework itself is standard: it is the McKendrick–von Foerster machinery of mathematical demography [4, 5], and age-structured variants of within-host models are well studied. The novelty here is not the framework. It is that the kernels I_{fix} and g are *derived in closed form from a mechanistic intracellular stochastic process*, rather than posited phenomenologically — together with the consequences drawn in the next three sections. Any honest write-up must say so plainly.

3.3 Numerical verification

Result 3.1 was verified end-to-end by a dedicated numerical suite (`verify_result_20_1.py`); the full catalogue is Appendix B. The tests cover four layers: kernel integrity (the inputs obey the single-cell identities, five parameter sets); the $\gamma T = 0$ reduction (with no reinfection, (26)–(27) collapse to $\mathcal{I} = \mathcal{I}_0 I_{\text{fix}}$ and the forced linear equation $\mathcal{V}' = \mathcal{I}_0 g - c\mathcal{V}$, matching the single-cell convolution to machine precision, and with $c = 0$ giving $\mathcal{V} = \mathcal{I}_0 V(t)$); structural limits (exponential kernels reproduce classical BMVR to relative error $< 5 \times 10^{-5}$, Figure 6; the late exponential growth rate equals the characteristic root r_* of §4 to $\sim 10^{-7}$, Figure 14a; the R_0 threshold is sharp, Figure 14b); and stochastic ground truth — an independent Gillespie ensemble of BDC cells with virion clearance reproduces the renewal means for $\mathcal{I}$ and $\mathcal{V}$ within sampling error ($\gamma T = 0$), Figure 7. **Headline: all 54 checks pass** (the full catalogue, with tallies and reproduction instructions, is Appendix B).

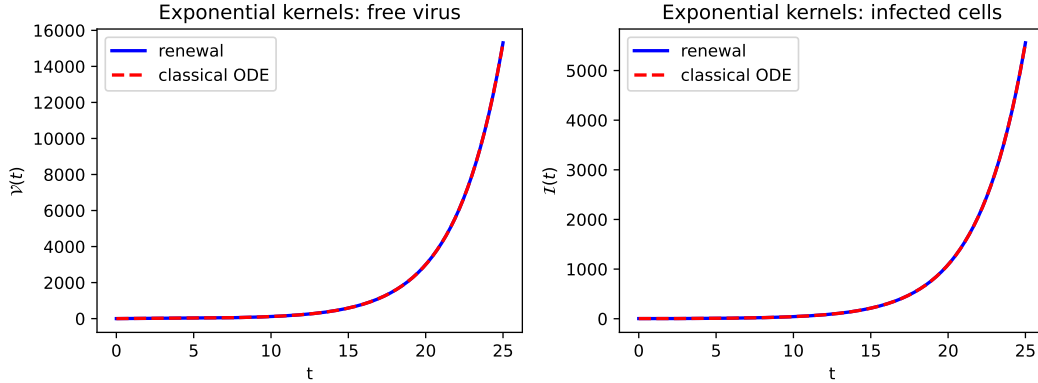


Figure 6: Structural correctness: when the kernels are pure exponentials, $S = e^{-da}$, $g = pe^{-da}$, the renewal system (26)–(27) is classical BMVR with those (p, d) . Relative discrepancy $< 5 \times 10^{-5}$ for three matched parameter pairs.

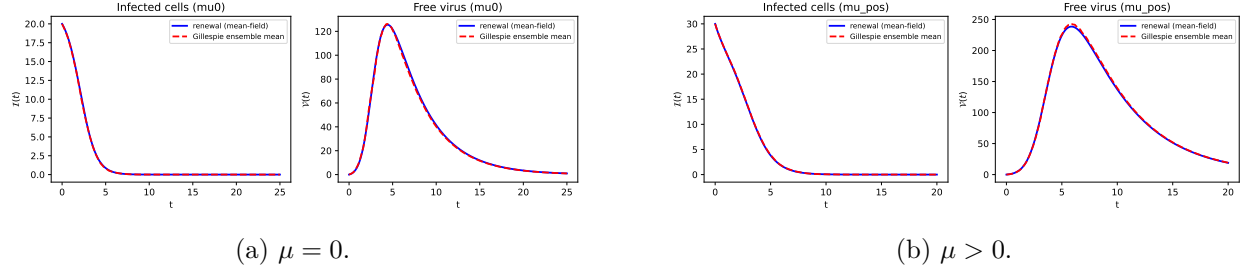


Figure 7: Stochastic ground truth: ensemble means of an independent Gillespie simulation of BDC cells (impulse releases at burst times, exponential clearance, no reinfection) against the renewal solution, relative $L^2 \sim 1\text{--}2\%$.

3.4 Overlays: classical vs renewal

To see what age structure does to an epidemic, integrate both systems side by side. In Figures 8–12, solid navy is the renewal model (26)–(27); dashed orange with circles is classical BMVR with the fairest constant-parameter match at the $r = 0$ end of the effective-rate map of §4,

$$p = p_{\text{eff}}(0) = \frac{V_{\infty}}{\mathbb{E}(T_{\text{prod}})} = \frac{\lambda V_{\infty}}{\log(a/(a-1))}, \quad d_{\mathcal{I}} = d_{\mathcal{I},\text{eff}}(0) = \frac{\lambda}{\log(a/(a-1))}, \quad (29)$$

so that total lifetime output and mean productive lifetime agree.

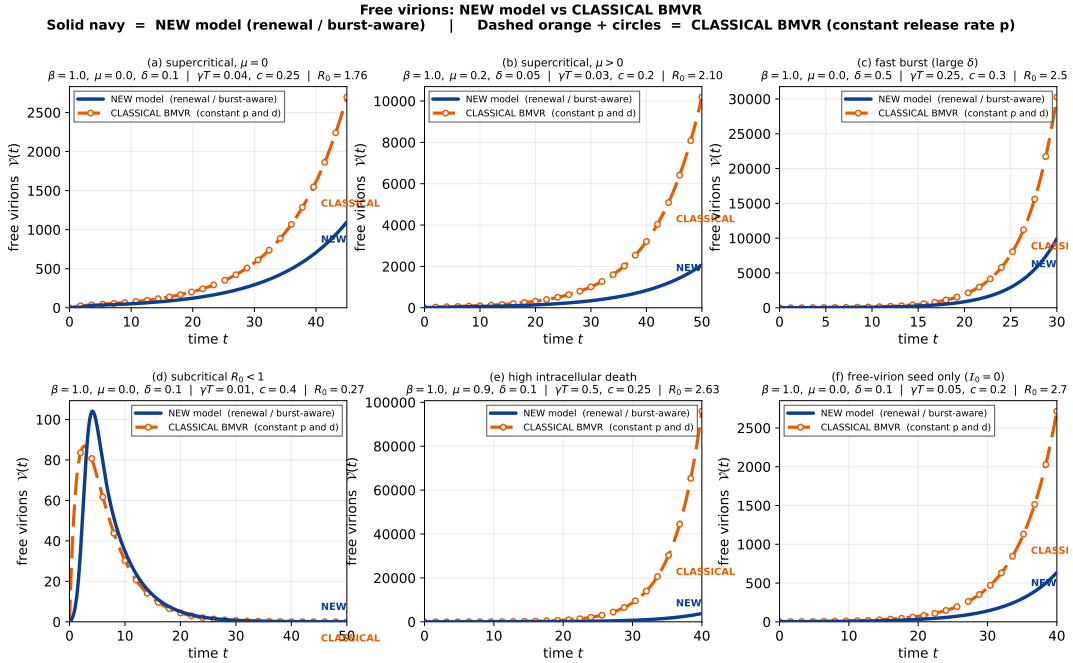


Figure 8: Free-particle trajectories $\mathcal{V}(t)$. Solid navy: renewal; dashed orange with circles: classical BMVR matched at $r = 0$. Six scenarios: supercritical growth, subcritical decay, fast burst, high intracellular death, and free-particle-only seeding.

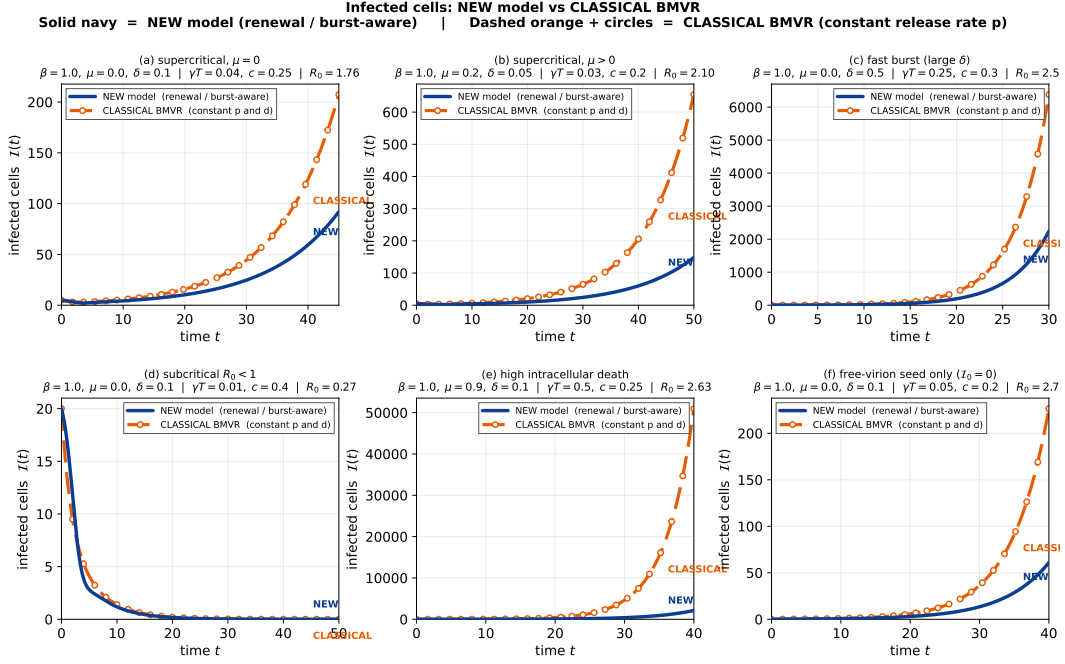


Figure 9: Infected-cell trajectories $\mathcal{I}(t)$ for the same six scenarios; same line code as Figure 8.

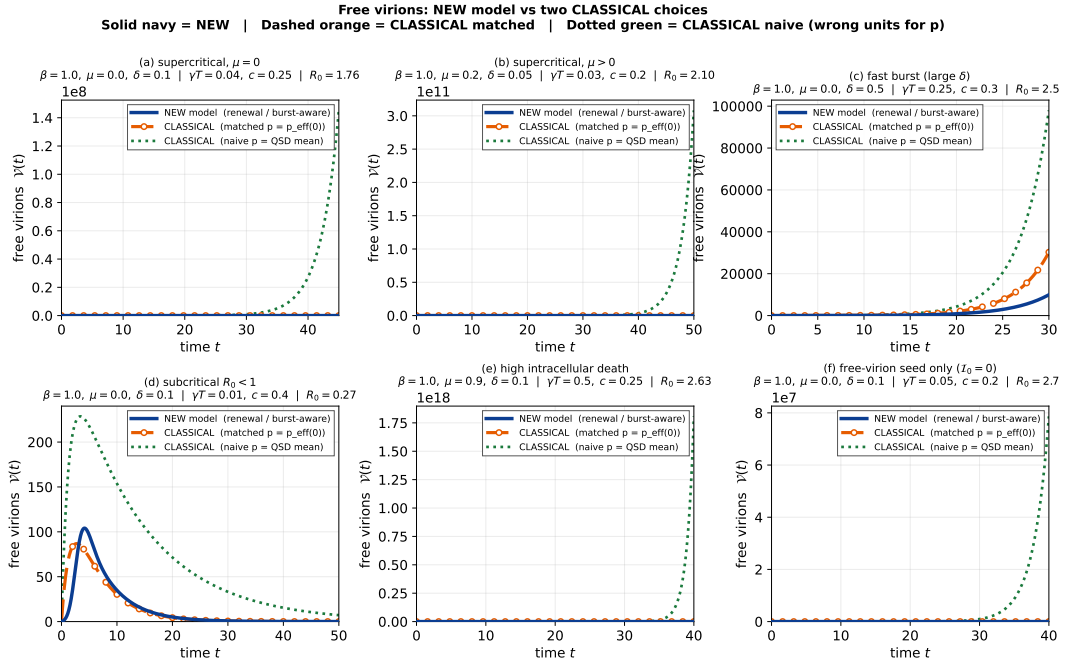


Figure 10: The naive proposal (21) as a third curve (dotted green): classical BMVR with $p = \langle X \rangle_{QS}$ — a count used as a rate — is systematically wrong in scale, not merely in shape.

The reading is consistent across panels. The matched classical model often gets the order of magnitude but not the shape: it misses the early lag (young cells dominate, and release is closer to

the young-cell limit $p \rightarrow \delta$ than to $p_{\text{eff}}(0)$), the timing of the peak, and the late growth rate (as the epidemic grows, generation-time structure moves the operative release rate away from its $r = 0$ value). The relative free-particle discrepancy,

$$\frac{|\mathcal{V}^{\text{new}}(t) - \mathcal{V}^{\text{classical}}(t)|}{\max_s \mathcal{V}^{\text{new}}(s)}, \quad (30)$$

is often order one or larger by the end of the plotted window in supercritical regimes, and is worst when intracellular death is high (Figure 11): there, V_∞ is small while the age-structure of release remains strongly non-exponential. Discrepancies grow with R_0 and with the strength of age-structure in the kernels (slow bursting, non-zero μ).

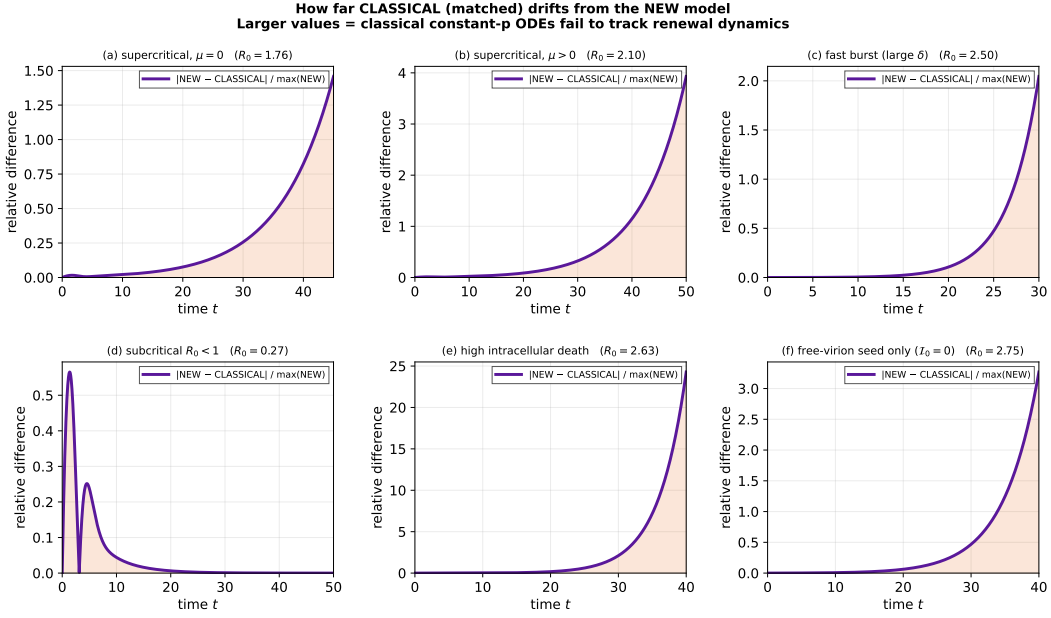


Figure 11: Relative free-particle discrepancy between the renewal model and matched classical BMVR across the six scenarios. Large values mean the constant- p ODEs fail to track the burst-aware model.

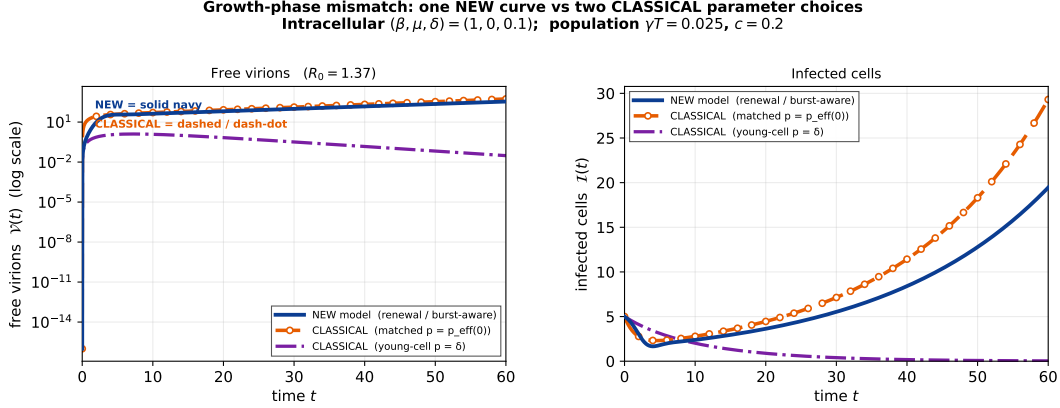


Figure 12: Growth-phase mismatch for one supercritical set ($\lambda = 1$, $\mu = 0$, $\delta = 0.1$). Solid navy: renewal; dashed orange: classical matched at $r = 0$; dash-dot purple: classical with the young-cell parameters $p = \delta$, $d_I = \mu + \delta$. Neither classical choice tracks the renewal curve across the whole trajectory — the $r = 0$ match fails early, the young-cell match fails late. Left panel logarithmic in V .

These numerical facts are exactly what the analytic reduction of the next section predicts: a constant pair (p, d_I) can reproduce one exponential growth rate exactly, but not a transient that samples a continuum of cell ages.

4 Effective parameters, R_0 , and identifiability

The renewal model (26)–(27) does not merely disagree with classical BMVR in transients; it contains classical BMVR as an exact special case, and the relationship between the two is precise enough to be turned into predictions.

4.1 Exact reduction in the exponential phase

Suppose the incidence grows exponentially, $i(t) = i_0 e^{rt}$, and look at times late enough that the lower limits of the convolutions are irrelevant. Then

$$\mathcal{I}(t) = i_0 e^{rt} \widetilde{I_{\text{fix}}}(r), \quad \mathcal{V}'(t) = i_0 e^{rt} \delta \widetilde{K}(r) - c \mathcal{V}(t), \quad (31)$$

where $\widetilde{f}(r) = \int_0^\infty e^{-r\alpha} f(\alpha) d\alpha$. A classical model $\mathcal{I}' = i - d_I \mathcal{I}$, $\mathcal{V}' = p \mathcal{I} - c \mathcal{V}$ driven by the same incidence has $\mathcal{I}(t) = i_0 e^{rt} / (r + d_I)$ and the eigenvalue relation $(r + d_I)(r + c) = \gamma T p$. Matching the two exponential solutions fixes

$$p_{\text{eff}}(r) = \frac{\delta \widetilde{K}(r)}{\widetilde{I_{\text{fix}}}(r)}, \quad d_{I,\text{eff}}(r) = \frac{1}{\widetilde{I_{\text{fix}}}(r)} - r. \quad (32)$$

This reduction is *exact*, not approximate: substituting (32) into $(r + d_I)(r + c) = \gamma T p$ returns

$$r + c = \gamma T \delta \widetilde{K}(r), \quad (33)$$

which is precisely the characteristic equation of the renewal system. Thus for every growth rate r there is a unique constant-parameter BMVR that reproduces the renewal model's exponential solution exactly. What (32) cannot do is reproduce transients: a single pair (p, d_I) matches one

growth rate, and a transient samples a continuum of them (Figure 6 shows the exponential case; the overlays of §3.4 show everything else).

The two ends of the map $r \mapsto (p_{\text{eff}}, d_{\mathcal{I},\text{eff}})$ are explicit. Using $\int_0^\infty \delta K = V_\infty$ (from $V' = \delta K$) and $\int_0^\infty I_{\text{fix}} = \mathbb{E}(T_{\text{prod}})$ ((13)),

$$r = 0 : \quad p_{\text{eff}}(0) = \frac{V_\infty}{\mathbb{E}(T_{\text{prod}})} = \frac{\lambda V_\infty}{\log(a/(a-1))}, \quad d_{\mathcal{I},\text{eff}}(0) = \frac{\lambda}{\log(a/(a-1))}; \quad (34)$$

the $r = 0$ value of p_{eff} coincides with naive moment-matching — lifetime output divided by lifetime — a pleasant consistency check, and the matching used in the overlays of §3.4. At the other end,

$$r \rightarrow \infty : \quad p_{\text{eff}}(r) \rightarrow \delta, \quad d_{\mathcal{I},\text{eff}}(r) \rightarrow \mu + \delta, \quad (35)$$

the *young-cell* limit: in a fast-growing epidemic almost every infected cell was infected moments ago, holds one particle, bursts at rate δ , and releases one particle when it does. Between the two ends, p_{eff} decreases monotonically in r across every parameter set we examined (Figure 13): fast epidemics are dominated by young cells and release slowly per cell; slow or stationary ones are dominated by old cells and release quickly. At $(\lambda, \mu, \delta) = (1, 0, 0.1)$: $p_{\text{eff}}(0) \approx 4.59$ against $p_{\text{eff}}(\infty) = 0.1$ — the same pathogen, two orders of magnitude apart, depending only on how fast the epidemic around it is growing.

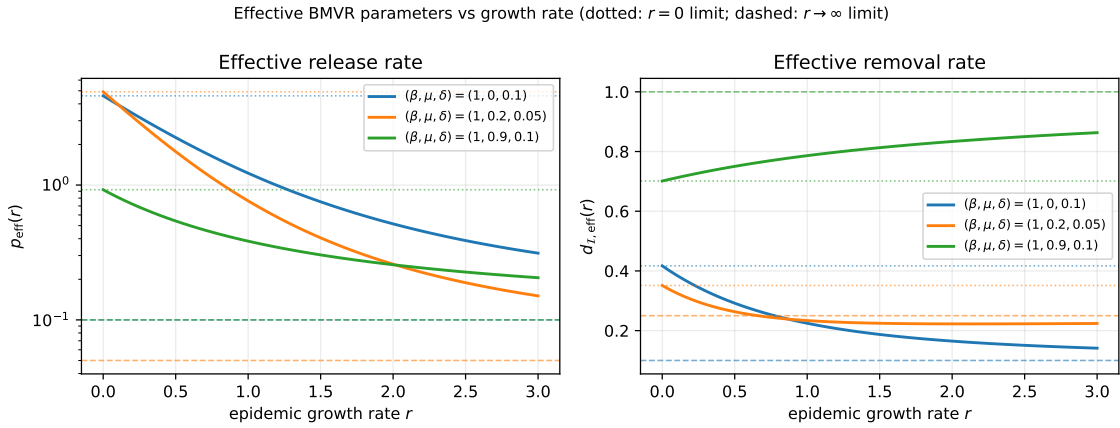


Figure 13: The effective release and removal rates (32) as functions of the epidemic growth rate r , for three parameter sets. Dotted and dashed lines mark the limits (34) and (35). p is not a constant of the pathogen; it is a function of the epidemic's phase.

The closed hypergeometric forms of $\widetilde{I}_{\text{fix}}$ and $\delta \widetilde{K}$ are given in Appendix C.1; the curves above are computed by direct quadrature.

4.2 The characteristic equation and the invariance of R_0

Setting $r = 0$ in (33) gives the threshold condition $\gamma T \delta \widetilde{K}(0) = c$, i.e. $\gamma T V_\infty = c$, so the basic reproduction number of the renewal model is

$$R_0^{\text{ODE}} = \frac{\gamma T V_\infty}{c}. \quad (36)$$

Theorem 4.1 (R_0 is unchanged by bursting). *At fixed total per-cell output V_∞ , replacing constant budding by bursting leaves R_0 exactly unchanged. Bursting alters the generation-time distribution — hence the growth rate r , hence the extinction probability, hence the stochastic early phase — but not the threshold itself.*

This is the classical R_0 -versus- r distinction in a setting where both sides are computable in closed form, and it connects the renewal model directly to the “perilous beginning” theme of Chapter 3 and to the flooding question of §5. Two cautions accompany it. First, the numerical checks of Figures 14a and 14b: the late growth rate of (26)–(27) agrees with the characteristic root r_* of (33) to $\sim 10^{-7}$ across five scenarios, and trajectories decay for $R_0 < 1$ and grow for $R_0 > 1$, the transition resolved already at $R_0 = 1.05$. Second, a notational care: (36) is the *continuum* reproduction number, in which one virion contributes mean $\gamma T/c$ new infections. A discrete branching process in which virions are absorbed on infection carries instead the mean $m = qV_\infty$, $q = \gamma T/(\gamma T + c)$; the two agree when $\gamma T \ll c$, and it is m that governs the branching calculations of §5. We write R_0 carefully in each context.

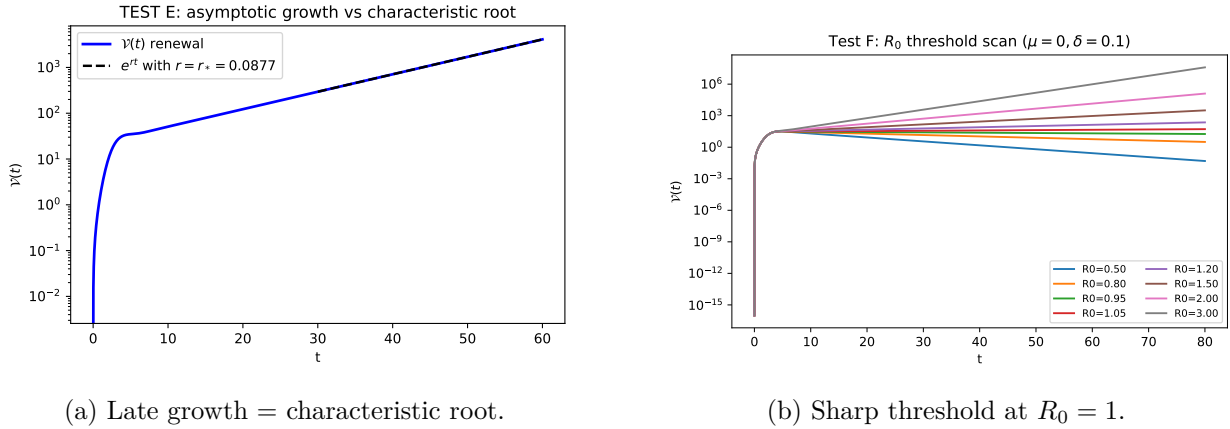


Figure 14: Two structural checks of the renewal system: the fitted late growth rate of $\mathcal{V}$ against the root of (33) (five scenarios, relative error $\sim 10^{-6}$ – 10^{-7}), and the R_0 threshold scan (eight values through $R_0 = 1$).

4.3 Identifiability: a negative result

How much of the intracellular process can a population-level fit recover? At any fixed growth rate r , the map

$$(\lambda, \mu, \delta) \mapsto (p_{\text{eff}}(r), d_{\mathcal{I}, \text{eff}}(r)) \quad (37)$$

sends three rates to two numbers, and is therefore not invertible.

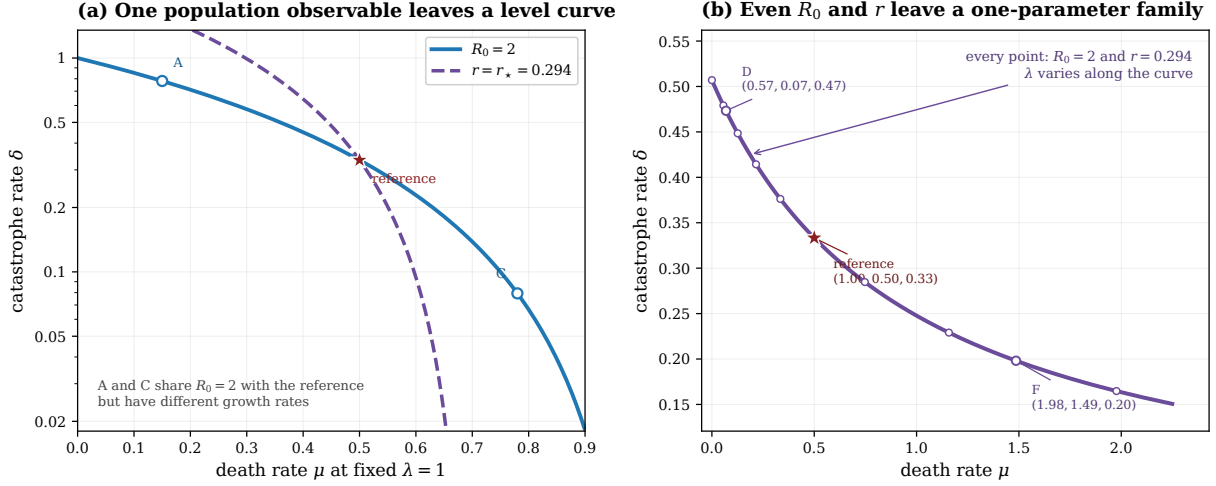


Figure 15: Population observables leave level sets rather than identifying the intracellular rates. We set $c = 1$ and $\gamma T = 1$, so that the interior flooding point $(\lambda, \mu, \delta) = (1, 0.5, 1/3)$ has $R_0 = 2$ and $r_* = 0.294$. (a) In the fixed slice $\lambda = 1$, $R_0 = 2$ and $r = r_*$ are distinct contours; the marked alternatives A and C share R_0 with the reference but not its growth rate. (b) Once λ is allowed to vary, a continuous family of distinct triples shares both $R_0 = 2$ and $r = 0.294$. Thus even joint knowledge of threshold and exponential growth does not identify (λ, μ, δ) .

This closes, negatively, the ambition of the proposal of §2.3, which hoped that a good estimate of p would grant access to the rates constituting the quasi-stationary mean: it would not, and neither would (p, d_I) together. A third observable is required. In order of experimental accessibility, the natural candidates are:

1. the burst-size dispersion — the geometric ratio $1/a$ fixes one combination of the rates directly (the burst-size law of Chapter 4a);
2. the fraction of cells that clear without bursting, which is b ;
3. the shape — not merely the mean — of the release-time density δJ (Chapter 4a).

Any one of these, together with (p, d_I) , over-determines (λ, μ, δ) and converts the fit into a consistency test of the whole intracellular model. The negative result is worth stating plainly: it is not a limitation of a fitting procedure but of the information content of constant-parameter BMVR itself.

4.4 Prediction for fitting practice

Since p_{eff} depends on r and decreases with it (Figure 13), BMVR fits to acute-phase data — fast growth, young cells — and to set-point data — slow or stationary — should yield systematically different estimates of p , in a direction and by an amount the theory specifies. Equivalently: the eclipse-like delay that within-host models usually insert by hand emerges here from the intracellular dynamics with no extra parameter at all — the renewal system's early lag, visible in Figures 8 and 12, *is* the eclipse.

5 The flooding advantage and the growth-rate trade-off

Does bursting help a pathogen get established? The question belongs to the stochastic early phase, where censuses are small and extinction is a real threat, and it can now be answered exactly: both the bursting and the budding version of the early infection are branching processes whose offspring distributions are available in closed form.

5.1 Offspring of one infected cell

In the branching approximation each released particle independently either infects a target cell (rate γT) or is cleared (rate c), so it infects with probability

$$q = \frac{\gamma T}{\gamma T + c}. \quad (38)$$

The offspring of one infected cell is the burst size $\mathcal{K}$, thinned by this success probability. The (unconditional) burst-size probability generating function is, summing the law (10), $G_{\mathcal{K}}(z) = b + (\delta/\lambda) z/(a - z)$ — including the mass b of cells that never burst — and thinning replaces z by $y := 1 - q + qz$:

$$G_{\text{off}}(z) = b + \frac{\delta}{\lambda} \frac{y}{a - y}, \quad y = 1 - q + qz. \quad (39)$$

One checks $G_{\text{off}}(1) = b + (\delta/\lambda)/(a - 1) = b + B = 1$. The mean and variance of the offspring number are

$$m = G'_{\text{off}}(1) = q V_{\infty}, \quad \text{Var}_{\text{burst}} = \frac{2q^2 V_{\infty}}{a - 1} + q V_{\infty} - q^2 V_{\infty}^2, \quad (40)$$

the variance following from $\text{Var}(\mathcal{K})$ via the limit of (12) and the standard composition formula for random sums. When $\gamma T \ll c$, $q \approx \gamma T/c$ and $m \approx R_0^{\text{ODE}}$ of (36).

5.2 Extinction probability in closed form

The extinction probability of a branching process is the smallest root of $G_{\text{off}}(z) = z$ in $[0, 1]$. With $y = 1 - q + qz$, the fixed-point equation is quadratic, $z = 1$ is always a root, and the other root gives, for $m > 1$,

$$z_{\text{ext}}^{\text{burst}} = \frac{a(1 - q + qb) - 1 + q}{q} = \frac{a - 1}{q} + 1 - a(1 - b). \quad (41)$$

5.3 Comparison with matched-mean budding: the main theorem

The comparator is the classical picture: a cell with $\text{Exp}(d_{\mathcal{I}})$ lifetime releasing at constant rate p . The number it releases is a Poisson process stopped at an independent exponential time, hence geometric on $\{0, 1, 2, \dots\}$ with mean $p/d_{\mathcal{I}}$; matched to the bursting cell's total output, $p/d_{\mathcal{I}} = V_{\infty}$. Thinning a geometric law leaves a geometric law, so the budding offspring number is geometric with the same mean $m = qV_{\infty}$, and its extinction probability is

$$z_{\text{ext}}^{\text{bud}} = \frac{1}{m} \quad (m > 1). \quad (42)$$

Theorem 5.1 (The flooding advantage). *Write $L := a(1 - b) = a - \mu/\lambda$, so that $V_{\infty} = L/(a - 1)$. For $m > 1$,*

$$z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L - 1) \left(\frac{1}{m} - 1 \right). \quad (43)$$

Hence bursting strictly lowers the establishment extinction probability, relative to matched-mean budding, if and only if $L > 1$, i.e.

$$\delta(\lambda + \mu) > \lambda\mu \quad \Longleftrightarrow \quad \boxed{\frac{1}{\delta} < \frac{1}{\lambda} + \frac{1}{\mu}} \quad (44)$$

(with the convention that the condition holds for all $\delta > 0$ when $\mu = 0$). When $L = 1$ the two extinction probabilities are identical for every q .

Proof sketch. The identity (43) is algebra: substitute (41), $m = qV_\infty = qL/(a - 1)$ and simplify. For the criterion, $L > 1$ is $a(1 - b) > 1$, i.e. $a > 1 + \mu/\lambda$; inserting the definition of a and squaring (both sides are positive, or the right-hand side is not) yields $(\lambda + \mu + \delta)^2 - 4\lambda\mu > 4\lambda^2(1 + \mu/\lambda - \eta)^2$, which reduces to $\delta(\lambda + \mu) > \lambda\mu$. $\square$

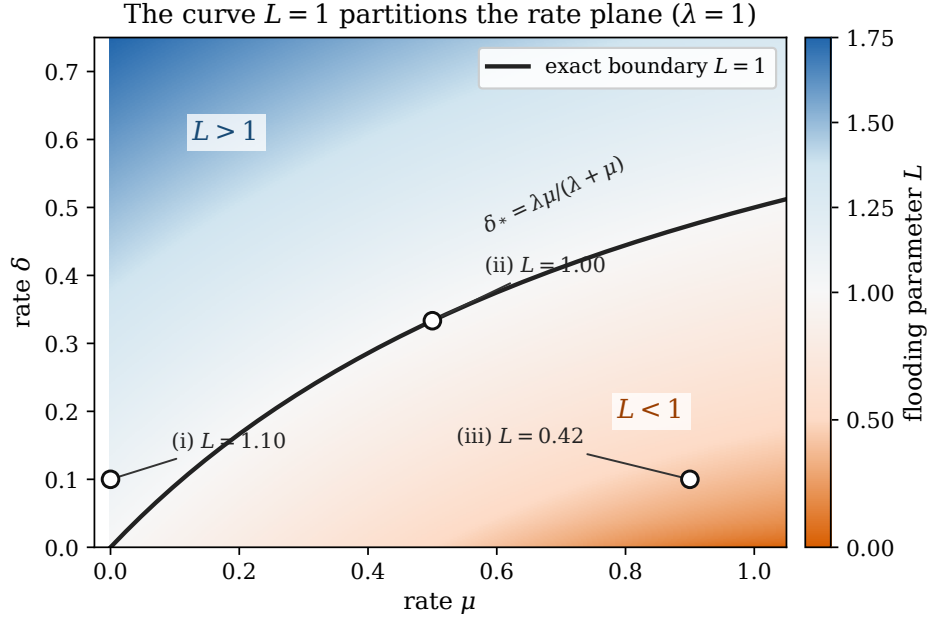


Figure 16: Flooding parameter $L(\mu, \delta)$ at $\lambda = 1$. The exact curve $L = 1$, equivalently $\delta_* = \lambda\mu/(\lambda + \mu)$, partitions the rate plane into $L > 1$ and $L < 1$. The three marked table points are $(\mu, \delta) = (0, 0.1)$, $(0.5, 1/3)$, and $(0.9, 0.1)$, for which $L = 1.10$, 1.00 , and 0.42 , respectively.

Numerically, the identity (43) is exact in every case we checked:

(λ, μ, δ)	q	$z_{\text{ext}}^{\text{burst}}$	$z_{\text{ext}}^{\text{bud}}$	verdict
$(1, 0, 0.1)$	0.2	0.400	0.455	bursting wins ($L = 1.10$)
$(1, 0.9, 0.1)$	0.9	0.935	0.844	budding wins ($L = 0.42$)
$(1, 0.5, 1/3)$	0.6	0.833	0.833	tie ($L = 1$)
$(1, 0.5, 1/3)$	0.9	0.556	0.556	tie ($L = 1$)

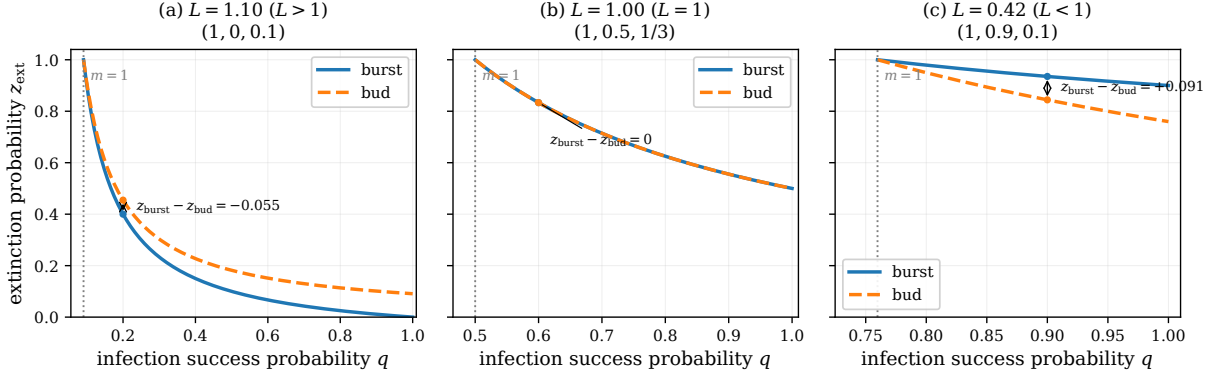


Figure 17: Extinction probabilities for bursting and matched-mean budding across the three flooding regimes. The curves begin at the supercritical boundary $m = 1$ and their signed gaps follow (43): bursting wins for $L > 1$, ties budding for $L = 1$, and loses for $L < 1$. From left to right, $(\lambda, \mu, \delta) = (1, 0, 0.1)$, $(1, 0.5, 1/3)$, and $(1, 0.9, 0.1)$.

For $m \leq 1$ both processes die out almost surely and the comparison is empty; (43) changes sign there, an artefact of both formulae running outside their domain $[0, 1)$, not a reversal of the biology.

The interpretation is that the flooding advantage is real but bought against a risk. Bursting concentrates output into one batch — lowering the offspring variance relative to the geometric output of a budding cell — but it carries the all-or-nothing hazard that the intracellular population dies out first, with probability b , and the cell releases nothing. The condition says the advantage survives exactly when the burst timescale $1/\delta$ is shorter than the harmonic-type combination $1/(1/\lambda + 1/\mu)$ of the birth and death timescales. When $\mu = 0$ the hazard of internal extinction vanishes and bursting always wins.

5.4 The boundary $L = 1$: a coincidence of laws

The boundary case is more than a tie of extinction probabilities.

Proposition 5.2. *The following are equivalent:*

$$L = 1 \iff a = 1 + \frac{\mu}{\lambda} \iff \delta(\lambda + \mu) = \lambda\mu \iff \frac{1}{\delta} = \frac{1}{\lambda} + \frac{1}{\mu}. \quad (45)$$

When they hold, the bursting offspring law (39) is exactly geometric with mean m — not merely equal in extinction probability to the budding comparator. In particular the offspring variances coincide, and

$$\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}} = \frac{2q^2 V_{\infty}(L - 1)}{a - 1}, \quad (46)$$

so that at matched mean m the bursting offspring variance is smaller than the budding one when $L > 1$, equal when $L = 1$, and larger when $L < 1$.

Proof. $L = 1$ is $a(1 - b) = 1$, i.e. $a - 1 = ab = \mu/\lambda$, the chain of equivalences above. Under it, $b = \delta/\lambda$ (by (3)), and (39) collapses:

$$G_{\text{off}}(z) = b \left(1 + \frac{y}{a - y} \right) = \frac{ab}{a - y} = \frac{a - 1}{a - y} = \frac{(a - 1)/a}{1 - y/a},$$

using $ab = a - 1$. With $y = 1 - q + qz$,

$$1 - \frac{y}{a} = \frac{a - 1 + q}{a} \left(1 - \frac{q}{a - 1 + q} z \right),$$

so

$$G_{\text{off}}(z) = \frac{1 - r}{1 - rz}, \quad r = \frac{q}{a - 1 + q} = \frac{m}{1 + m}, \quad m = \frac{q}{a - 1} = qV_{\infty},$$

which is the probability generating function of a geometric law on $\{0, 1, 2, \dots\}$ with mean m — the same law as the matched budding comparator. The variance identity (46) is the difference between the geometric variance $m(1 + m)$ and (40), with $V_{\infty} = L/(a - 1)$. $\square$

At the boundary, then, bursting and budding are the same branching process in different clothes. Away from it, (46) quantifies the direction in which they differ.

5.5 The deterministic growth-rate trade-off

Establishment is only half of the story; the other half is how fast an established infection grows. Match the two models a second time, now at level R_0 and at mean productive lifetime: take the budding cell with $d_{\mathcal{I}} = 1/\mathbb{E}(T_{\text{prod}})$ and $p = V_{\infty}d_{\mathcal{I}}$ — which is exactly $p_{\text{eff}}(0)$, so this is the fairest possible classical match — and choose γT (or c) so that both models share $R_0 = 2$. Then solve for the deterministic growth rates: r_{bud} from the quadratic $(r + d_{\mathcal{I}})(r + c) = \gamma T p$, and r_{burst} from the characteristic equation (33). In every regime we checked — $L > 1$, $L = 1$, and $L < 1$ — the ordering is the same:

$$r_{\text{bud}} > r_{\text{burst}}. \tag{47}$$

With $c = 1$ and $R_0 = 2$:

(λ, μ, δ)	L	$d_{\mathcal{I}}$	r_{bud}	r_{burst}
(1, 0, 0.1)	1.10	0.417	0.250	0.180
(1, 0.5, 1/3)	1.00	0.910	0.395	0.294
(1, 0.9, 0.1)	0.42	0.701	0.343	0.198

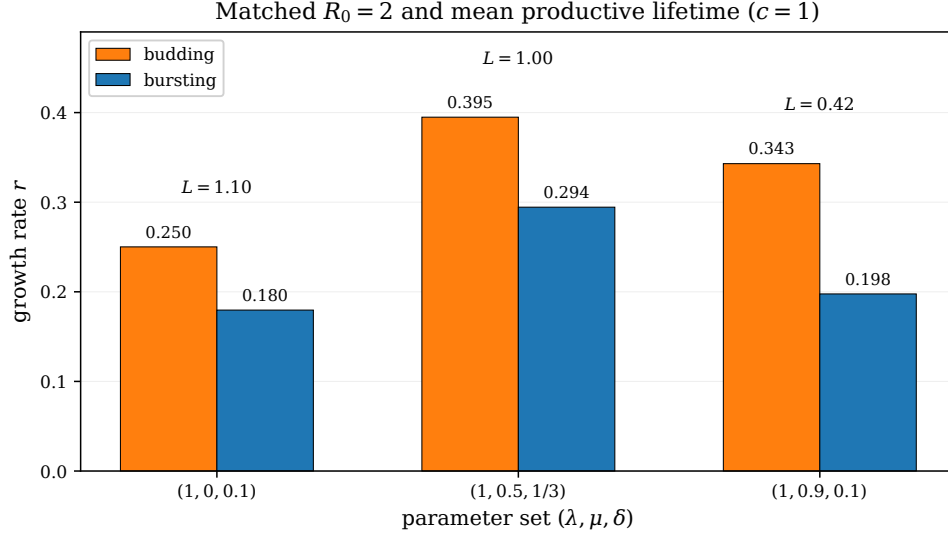


Figure 18: Deterministic growth rates at matched $R_0 = 2$, clearance rate $c = 1$, and mean productive lifetime. Budding invades faster than bursting in all three flooding regimes. The parameter sets $(\lambda, \mu, \delta) = (1, 0, 0.1)$, $(1, 0.5, 1/3)$, and $(1, 0.9, 0.1)$ have $L = 1.10$, 1.00 , and 0.42 , respectively.

The mechanism is generation time: the budding cell starts releasing immediately, while the bursting cell must first accumulate load, so budding's generation times are shorter and its exponential invasion is faster — the same delay that appeared as the eclipse-like lag of Figures 8 and 12.

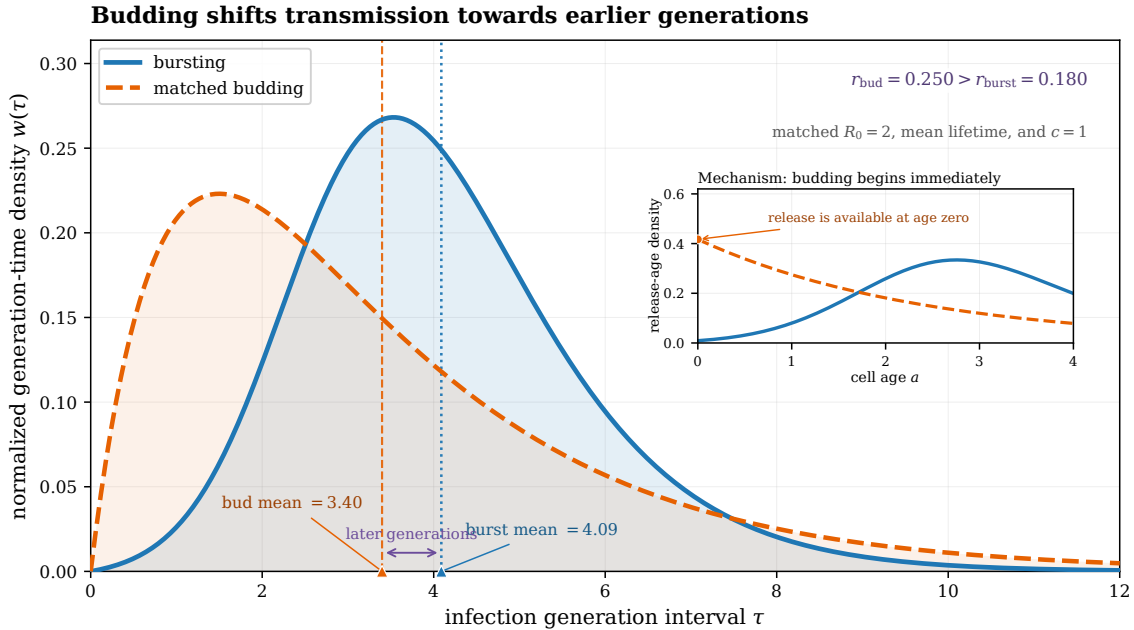


Figure 19: Generation-time structure behind the first row of Figure 18. For $(\lambda, \mu, \delta) = (1, 0, 0.1)$, budding and bursting are matched at $R_0 = 2$, $c = 1$, total yield, and mean productive lifetime. The main curves are the normalized infection-generation laws, including the common extracellular waiting-time component; the inset isolates release age and shows why the budding law leads. Its mean generation time is 3.398 against 4.089 for bursting, consistent with $r_{\text{bud}} = 0.250 > r_{\text{burst}} = 0.180$.

The two comparisons together give the biological headline. At fixed total output per cell, bursting can lower the probability of early extinction (when $L > 1$) while simultaneously slowing deterministic invasion speed: R_0 stays fixed, and z_{ext} and r move in opposite directions. A pathogen tuning its burst rate δ faces a genuine trade-off. For $\mu > 0$, budding beats bursting on extinction until δ crosses the threshold $\delta_* = \lambda\mu/(\lambda + \mu)$ — the $L = 1$ boundary — beyond which bursting takes over; for $\mu = 0$ bursting wins on extinction at every δ , at the continuing cost of a slower r .

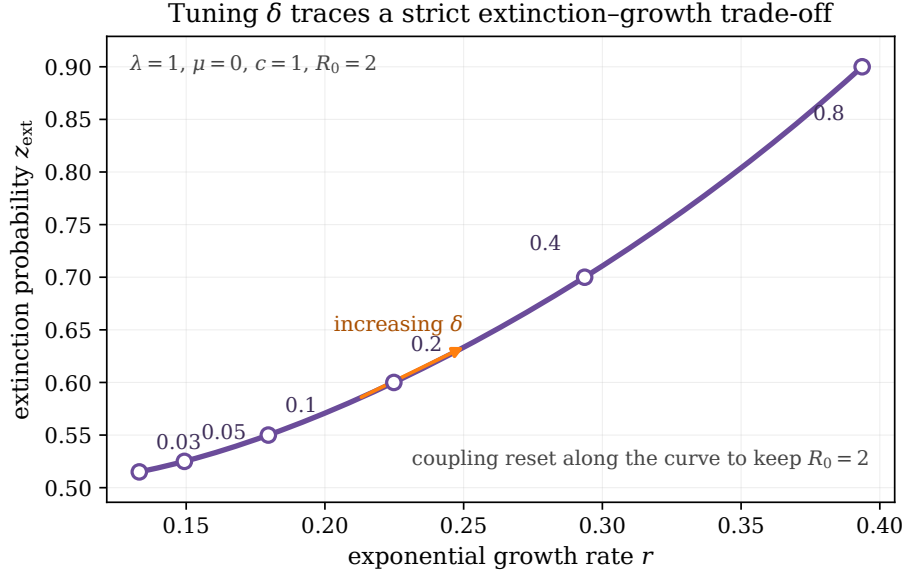


Figure 20: Parametric extinction–growth trade-off as δ varies from 0.03 to 0.8, with $\lambda = 1$, $\mu = 0$, and clearance rate $c = 1$. Matching protocol: for each δ , the coupling is reset to fix $R_0 = 2$; the constant-production comparator is then matched in mean yield V_∞ and mean producer lifetime $\mathbb{E}[T]$ through $d = 1/\mathbb{E}[T]$ and $p = V_\infty d$. Increasing δ raises both r and z_{ext} , so faster growth is paid for by a higher extinction probability.

Remark 5.3 (Scope of the comparison). Both results compare branching structures at fixed offspring mean. They say nothing about nonlinear effects — immune saturation, local depletion of targets, Allee-type cooperation — where other advantages of flooding may live; any full account should name and separate the two mechanisms. The evolutionary framing, and where this result sits relative to the bursting–budding spectrum of Chapter 6, is taken up there.

6 The spectrum of release models

Bursting and constant budding are the two ends of a spectrum, and the renewal construction of §3 is the bridge between them: *any* intracellular model that supplies a survival kernel $S(a)$ and a release kernel $g(a)$ induces the same population skeleton. This section lays out the spectrum. First the skeleton and the classical models built on it (§6.1–§6.2); then reset catastrophe, which interpolates by letting one cell release many times (§6.3); then self-exciting release (§6.4); and finally partial release, which interpolates in *how much* leaves the cell at an event (§6.5). The evolutionary comparison between the ends of the spectrum was made in §5; its ecological context belongs to Chapter 6. The whole arrangement is sketched in Figure 21; Figure 22 gives the same mechanisms ordered by their generation-time structure, with the multi-stage contrast class of §7 attached.

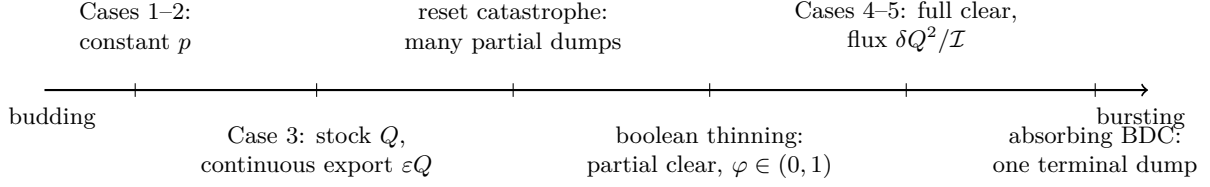


Figure 21: The budding–bursting spectrum. Moving right, release becomes more concentrated in time and more load-dependent: from constant budding (Cases 1–2) through stock-and-export (Case 3), recurrent partial dumps (reset catastrophe), a tunable fraction φ per event (boolean thinning), load-proportional full clear (Cases 4–5), to the single terminal dump of the absorbing BDC of this chapter.

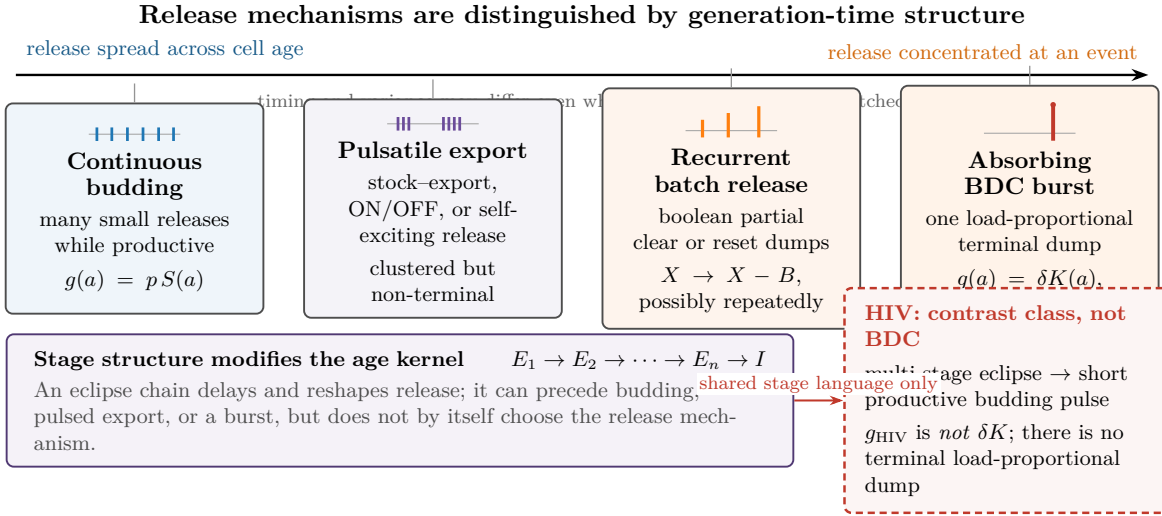


Figure 22: Spectrum of release mechanisms ordered by generation-time structure, from continuous production through pure bursting and modular variants. Multi-stage compartmental release models are shown as a contrast class, not as instances of the absorbing BDC process of this chapter.

6.1 The universal renewal skeleton

With incidence $i(t) = \gamma T \mathcal{V}(t)$ (or a nonlinear incidence $f(\mathcal{V}, T)$), any choice of kernels gives

$$\mathcal{I}(t) = \mathcal{I}_0 S(t) + (i * S)(t), \quad \frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g(t) + (i * g)(t) - c \mathcal{V}(t), \quad (48)$$

with effective parameters $p_{\text{eff}}(r) = \tilde{g}(r)/\tilde{S}(r)$ and, for linear incidence, $R_0 = (\gamma T/c) \int_0^\infty g$. The models of this section differ only in their (S, g) :

Model	$S(a)$	$g(a)$
Classical BMVR	$e^{-d_{\mathcal{I}}a}$	$p e^{-d_{\mathcal{I}}a}$
Absorbing BDC (this chapter)	$I_{\text{fix}}(a)$	$\delta K(a)$
MOI- k BDC (Chapter 4a)	$I(a)^k - D(a)^k$	$\delta K_k(a)$
Reset catastrophe (§6.3)	$e^{-d_{\mathcal{I}}a}$	$\delta e^{-d_{\mathcal{I}}a} \mathbb{E}(X_a^2)$
HIV multi-stage (§7)	stage survival	$p \times$ productive occupancy
Two-type / GATE (Ch. 5)	two-type I_{fix}	two-type second-moment flux

The first row is the check of Figure 6: exponential kernels reduce (48) exactly to classical BMVR.

6.2 Five modular ODE cases

When a finite-dimensional ODE description is wanted — for fitting, or for analytic tractability — the same ideas assemble into a ladder of five cases. Target cells T are held fixed throughout; $\mathcal{I}$ denotes productively infected cells, $\mathcal{V}$ free particles, E eclipse cells, and Q the total intracellular stock summed over infected cells. In all five cases, production into the stock (once productive) is *mean-linear*, immigration-type, at rate ν per cell — the choice appropriate to template-limited intracellular accumulation (§7.4); what distinguishes the cases is the release law and the presence or absence of an eclipse class. In this subsection α denotes the eclipse conversion rate; it is not the root a .

6.2.1 Case 1 — Classic BMVR

An infected cell produces particles that are released immediately at constant rate p (budding). No eclipse, no stored pool; infected cells are removed at rate $d_{\mathcal{I}}$:

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}. \quad (49)$$

Production and export are collapsed into the single phenomenological parameter p . This is the model the rest of the chapter is replacing.

6.2.2 Case 2 — Eclipse + classic release

New infections enter an eclipse class E — infected but not yet producing — and convert to productive cells at rate α (mean eclipse length $1/\alpha$), with optional eclipse death d_E :

$$\frac{dE}{dt} = \gamma T \mathcal{V} - (\alpha + d_E) E, \quad \frac{d\mathcal{I}}{dt} = \alpha E - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad (50)$$

$$R_0 = \frac{\gamma T}{c} \cdot \frac{\alpha}{\alpha + d_E} \cdot \frac{p}{d_{\mathcal{I}}}. \quad (51)$$

A fraction $\alpha/(\alpha + d_E)$ of eclipse cells become productive; each then yields mean $p/d_{\mathcal{I}}$ particles. Setting $\alpha \rightarrow \infty$ with $d_E = 0$ recovers Case 1. A more realistic eclipse-time distribution is an Erlang chain $E_1 \rightarrow \dots \rightarrow E_n \rightarrow \mathcal{I}$, each step at rate α ; the $\mathcal{V}$ equation is unchanged. This is the standard delay construction of the within-host literature — and, as §7 will emphasise, it is inserted by hand there, whereas in the two-type BDC it emerges mechanically (the GATE observation of Chapter 5).

6.2.3 Case 3 — Intracellular stock + continuous export

Productively infected cells accumulate stock at rate ν per cell; each intracellular unit is exported at per-capita rate ε ; cell death removes the cell and its share of stock:

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \varepsilon Q - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \varepsilon Q - c \mathcal{V}. \quad (52)$$

The mean load per productive cell is $m = Q/\mathcal{I}$. In the quasi-steady stock regime ($\frac{dQ}{dt} \approx 0$),

$$Q_* = \frac{\nu}{\varepsilon + d_{\mathcal{I}}} \mathcal{I}, \quad p_{\text{eff}} := \frac{\varepsilon Q_*}{\mathcal{I}} = \frac{\varepsilon \nu}{\varepsilon + d_{\mathcal{I}}}, \quad (53)$$

and Case 3 collapses to Case 1 with $p = p_{\text{eff}}$: *classical BMVR is the fast-export limit of stock-and-export dynamics*. When export is not fast, $\mathcal{V}$ lags and depends on the history of Q . With eclipse stacked in front (incidence into E , conversion at α , only $\mathcal{I}$ producing), the Q and $\mathcal{V}$ equations are unchanged and eclipse cells contribute nothing until conversion. The reproduction number is

$$R_0 = \frac{\gamma T \nu \varepsilon}{c d_{\mathcal{I}} (\varepsilon + d_{\mathcal{I}})} \quad (\text{no eclipse}), \quad (54)$$

which is Case 1's R_0 with $p = \varepsilon \nu / (\varepsilon + d_{\mathcal{I}})$.

6.2.4 Case 4 — Intracellular stock + proportional full clear

The same accumulation, but release now comes in load-proportional dumps: in a cell with load x , dumps occur at rate δx and the whole pool is transferred (the cell survives the dump; death remains the separate rate $d_{\mathcal{I}}$). Under the homogeneous-load closure $x \approx m = Q/\mathcal{I}$, the population dump flux is $\delta \mathbb{E}(\sum_i X_i^2) \approx \delta \mathcal{I} m^2 = \delta Q^2/\mathcal{I}$, giving

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \delta \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \delta \frac{Q^2}{\mathcal{I}} - c \mathcal{V}, \quad (55)$$

with the convention that the nonlinear terms vanish when $\mathcal{I} = 0$. The dump term is simultaneously export into $\mathcal{V}$ and clearance of Q ; it is quadratic in total load at fixed $\mathcal{I}$, i.e. superlinear in mean load — the mean-field echo of the superlinearity of g_k in Chapter 4a ((15)). The mature quasi-steady balance, $\nu = \delta m^2 + d_{\mathcal{I}} m$, gives

$$m = \frac{-d_{\mathcal{I}} + \sqrt{d_{\mathcal{I}}^2 + 4\delta\nu}}{2\delta}, \quad p_{\text{eff}} = \delta m^2, \quad (56)$$

which plays the role of p only in that steady regime: transients and early infection need the full $(Q, \mathcal{V})$ system. For invasion analysis near $\mathcal{I} = 0$, the map $Q \mapsto Q^2/\mathcal{I}$ is singular; one should linearise the underlying single-cell process instead, taking $R_0 = \gamma T V_{\infty}/c$ as in the renewal framework.

6.2.5 Case 5 — Eclipse + stock + proportional full clear

Case 2's eclipse block in front of Case 4: incidence creates eclipse cells; only productive cells accumulate stock and dump:

$$\frac{dE}{dt} = \gamma T \mathcal{V} - (\alpha + d_E) E, \quad \frac{d\mathcal{I}}{dt} = \alpha E - d_{\mathcal{I}} \mathcal{I}, \quad (57)$$

$$\frac{dQ}{dt} = \nu \mathcal{I} - \delta \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \delta \frac{Q^2}{\mathcal{I}} - c \mathcal{V}. \quad (58)$$

Eclipse adds a further staged lag — E rises first, then $\mathcal{I}$ and Q , then $\mathcal{V}$ — which Case 4 cannot produce; the quasi-steady relations for the mean load among productive cells are unchanged. In the single-cell sense,

$$R_0 = \frac{\gamma T}{c} \cdot \frac{\alpha}{\alpha + d_E} \cdot V_{\infty}. \quad (59)$$

6.2.6 Summary of the five cases

Case	Intracellular story	Free-particle source term
1 Classic	none (instant release)	$p \mathcal{I}$
2 Eclipse + classic	delay $E \rightarrow \mathcal{I}$, then as 1	$p \mathcal{I}$
3 Stock + continuous export	ν into Q , export εQ	εQ
4 Stock + full clear	ν into Q , dump $\propto Q^2/\mathcal{I}$	$\delta Q^2/\mathcal{I}$
5 Eclipse + stock + full clear	$E \rightarrow \mathcal{I}$, then as 4	$\delta Q^2/\mathcal{I}$

What is fixed across Cases 3–5 is the mean-linear production; what is chosen is the release law and the presence of eclipse, and Case 5 is exactly Case 4 composed with Case 2’s eclipse block. Each case also induces kernels (S, g) , so the convolution form (48) remains available whenever the ODE stage structure is replaced by age structure — which is precisely the absorbing BDC at the bursting end of the ladder.

6.3 Reset catastrophe

The absorbing BDC releases at most once per cell. An intermediate worth isolating lets a cell release *many* times: while the cell is alive, the load undergoes birth λn , death μn , optional immigration ν , and a *dump that resets*: at rate δn the whole pool n is released and the load returns to 0, after which it may re-accumulate. The cell itself dies at a separate rate $d_{\mathcal{I}}$, independent of the load. Biologically, read the load as a local assembled pool rather than the whole-cell contents: full dumping is one mathematical extreme of batch release, budding is many small exports, and this process is a tractable multi-pulse caricature between them.

Because dumps do not absorb, the kernels change but the skeleton does not:

$$S(a) = e^{-d_{\mathcal{I}} a}, \quad g(a) = \delta e^{-d_{\mathcal{I}} a} \mathbb{E}(X_a^2), \quad (60)$$

where $\mathbb{E}(X_a^2)$ is computed for the non-absorbing reset process. Linear rates keep the moments of that process tractable (closed ODEs under the generator), so g is explicit once they are solved; structurally, $g \propto \mathbb{E}(X^2)$ exactly as for the absorbing BDC. The renewal BMVR is identical in form to (26)–(27), with

$$p_{\text{eff}}(r) = (r + d_{\mathcal{I}}) \tilde{g}(r), \quad d_{\mathcal{I}, \text{eff}} \equiv d_{\mathcal{I}}, \quad (61)$$

and $R_0 = (\gamma T/c) \int_0^\infty g$ — finite only for $d_{\mathcal{I}} > 0$, since otherwise recurrent dumps could continue forever. If the reset process admits a stationary law π and $d_{\mathcal{I}}$ is small, the mature-cell limit is

$$p_{\infty} = \lim_{a \rightarrow \infty} \frac{g(a)}{S(a)} = \delta \mathbb{E}_{(\cdot)} \pi[X^2] : \quad (62)$$

a long-lived infected cell behaves as classical constant- p BMVR with that p , while transients (first dumps, young cells) still need the full kernel. An eclipse is added exactly as before: set $g(a) = 0$ for $a < \tau_e$ and keep $S(a) = e^{-d_I a}$.

	Absorbing BDC	Reset catastrophe
After dump	process ends in H	$X \rightarrow 0$, continues
Dumps per cell	≤ 1	many, until cell death
Survival kernel	I_{fix}	$e^{-d_I a}$
Release kernel	$\delta K_{\text{BDC}}(a)$	$\delta e^{-d_I a} \mathbb{E}(X_a^2)$
Population equations	renewal BMVR	same form
Mature limit	not constant budding	$p \rightarrow \delta \mathbb{E}(\cdot) \pi[X^2]$

6.4 Self-exciting release

A further biological possibility is that release is partially *self-exciting*: one event makes further release more likely for a short time — cooperative nucleation, local membrane platforms, ESCRT machinery already recruited at a site. The possibility must be balanced against depletion of stock and finite machinery: pure open-ended self-excitation is implausible, and a hybrid of excitation, refill and depletion is the reasonable picture. Two levels of model are available, as throughout: a renewal BMVR exists whenever the single-cell model defines (S, g) ; a finite ODE BMVR needs the excitation carried by a low-dimensional Markov state.

The menu of candidates:

#	Model	Excitation carried by	ODE BMVR?	Renewal?
1	Classical Hawkes	event history only	poor	yes (numerical g)
2	Hawkes + stock X	events + depletion/refill	via Model 10	yes
3	Marked Hawkes	size-weighted events	approximate	yes (numerical)
4	ETAS-style	size + time kernel	approximate	yes (numerical)
5	Cox + latent Z	latent excitability	if Z simple	yes
6	ON/OFF telegraph	sustained high phase	yes	yes
7	$\lambda \propto X^2$	current load only	yes (Cases 4–5)	yes
8	Nucleation biophysics	assembly cascade	hard	yes (numerical)
9	Age since last release	renewal excitation	yes (stages)	yes
10	Jumping rate $r(t)$	shot-noise intensity	yes (mean-field)	yes
11	Spatial Hawkes	local membrane	average first	after coarse-graining
12	Full clear (Cases 4–5)	instant total cascade	yes	yes

They nest: Model 2, with intensity $\lambda(t) = f(X_t) + \sum_{t_i < t} h(t - t_i)$, is the general event-history form — $h \equiv 0$ and $f(x) = \varepsilon x$ recovers continuous export (Case 3); $h \equiv 0$, $f(x) = \delta x^2$ with full clear recovers Model 7 and Cases 4–5; an exponential $h(t) = \eta e^{-\zeta t}$ is equivalent to Model 10; Model 6 is a coarse two-level version without a continuous intensity. The four focal models:

Model 7 — cooperative intensity, $\lambda(x) = \delta x^2$ (or $\delta x(x - 1)$). No memory of past events, only the current load: more material makes release disproportionately more likely. The single-cell process is production ν , optional degradation μX , and full-clear dumps at rate δX^2 ; the ODE form is exactly Cases 4–5 under the homogeneous closure, and the renewal form has $g(a) = \delta \mathbb{E}(X_a^2 \mathbf{1}\{\text{alive}\})$. Cooperativity without history; the default bulk “excitatory” model.

Model 6 — ON/OFF telegraph. The cell (or a membrane site) switches between OFF and ON at rates $\sigma_{\text{on}}, \sigma_{\text{off}}$; while ON it releases at rate ρ (or εX). Sample paths are intermittent surges without full clear. The simplest ODE version splits the infected class:

$$\frac{dI_{\text{OFF}}}{dt} = \gamma T \mathcal{V} + \sigma_{\text{off}} I_{\text{ON}} - (\sigma_{\text{on}} + d_{\mathcal{I}}) I_{\text{OFF}}, \quad (63)$$

$$\frac{dI_{\text{ON}}}{dt} = \sigma_{\text{on}} I_{\text{OFF}} - (\sigma_{\text{off}} + d_{\mathcal{I}}) I_{\text{ON}}, \quad \frac{d\mathcal{V}}{dt} = \rho I_{\text{ON}} - c \mathcal{V}, \quad (64)$$

and quasi-steady switching yields effective budding at the duty cycle, $p_{\text{eff}} = \rho \sigma_{\text{on}} / (\sigma_{\text{on}} + \sigma_{\text{off}})$ — a transparent limit back to constant p . Clustered release with writable ODEs.

Model 10 — jumping continuous release rate. Keep a continuous intensity $r(t) \geq 0$ that jumps up by η at each release event and decays to baseline r_0 at rate ζ : $dr = -\zeta(r - r_0)dt + \eta dN_t$, with flux $r(t)$ or $r(t)X_t$. This is the Markovian (shot-noise) form of Hawkes with exponential kernel. Tracking $\mathcal{I}$, total load Q and total excitation $R = \sum r_i$ ($\bar{r} = R/\mathcal{I}$), a mean-field closure gives ODEs for $(\mathcal{I}, Q, R, \mathcal{V})$ with flux $\approx \bar{r}Q$; true fading-memory self-excitation with finite-dimensional state, at the cost of closures and three extra parameters.

Model 2 — Hawkes with stock. The literal version: $\lambda(t) = f(X_t) + \sum_{t_i < t} h(t - t_i)$; on an event X drops by the released amount; between events X refills at rate ν ; the cell dies at rate $d_{\mathcal{I}}$. The renewal form is always well-defined ($g(a) = \mathbb{E}(\lambda(a) \mid \text{alive})$), typically numerical; an ODE form exists only after reduction to exponential h , i.e. to Model 10. The right language for “Hawkes-like budding” in a paper; the right implementation is Model 10.

	7 $\lambda \propto X^2$	6 ON/OFF	10 jumping r	2 Hawkes+ X
Carrier	load only	discrete phase	continuous r	event history (+load)
Memory	Markov in X	Markov in phase	Markov in r	non-Markov unless exp. h
Clusters	high $X \rightarrow$ fast dumps	long ON sojourn	high- r window	aftershocks
ODE BMVR	yes (Cases 4–5)	yes (split $\mathcal{I}$)	yes (mean-field R)	only via 10
Renewal	yes	yes	yes	yes
$\rightarrow$ classic p	QSS δm^2	duty cycle	QSS $\mathbb{E}(rX)$	as 10 if exp. h

Biologically, cooperative assembly makes short clustered release a medium–high confidence hypothesis; a bare Hawkes process with event memory and no depletion is a low–medium confidence standalone cell model. The practical ranking: Model 7 for writable ODEs inside Cases 3–5; Model 6 for intermittent release with the simplest new ODEs; Model 10 for fading memory with ODEs; Model 2 as the conceptual definition.

6.5 Boolean and partial release

The last axis of the spectrum is *how much* leaves the cell when a release event fires. Cases 3–5 sit at two extremes — continuous export vs full clear — and a natural intermediate removes only a random part of the load: at load n , release B with $0 \leq B \leq n$, retain $n - B$. Two ingredients are free: when events fire (rate $\lambda(X)$: constant δ , linear δX , or cooperative δX^2), and the law of $B \mid X = n$. The options:

- A. **Boolean thinning (recommended default):** each unit is released independently with probability φ , $B \mid X = n \sim \text{Binomial}(n, \varphi)$; $\varphi \rightarrow 0$ tiny fractional clear, $\varphi = 1$ full clear. Generating functions stay friendly.
- B. Fixed fraction $B = \text{round}(fn)$: same mean as A with $\varphi = f$, less stochasticity.
- C. Random fraction ($U \sim \text{Beta}$ or uniform thinning): more variable bursts than B.
- D. Fixed capped batch $B = \min(k, n)$: $k = 1$ is unit budding in the frequent-event limit; large k approaches full clear.
- E. Size-biased release, $P(B = j \mid n) \propto j$: natural if one site releases a random assembled blob.
- F. Threshold-then-partial: events only for $X \geq x_*$, then A or B — accumulation with a trigger.
- G. Short high-export window: a triggered burst of high ε for a brief time — partial clear in expectation, related to Model 6.

The single-cell boolean process: produce at rate ν ; release events at rate $\lambda(X)$ with $B \sim \text{Binomial}(X, \varphi)$ released and $X \rightarrow X - B$; cell death at d_I . Its nested limits:

Choice	Recovers
$\varphi = 1, \lambda = \delta X^2$	full clear, cooperative (Case 4 style)
$\varphi = 1, \lambda = \delta X$	full clear, linear hazard
$0 < \varphi < 1, \lambda = \delta X$	the main partial-clear intermediate
$B \equiv 1, \lambda = \varepsilon X$	unit budding (Case 3, discrete)

At the mean-field level with $\lambda = \delta X$, the population equations are Case 4 scaled by φ :

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_I \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \delta \varphi \frac{Q^2}{\mathcal{I}} - d_I Q, \quad \frac{d\mathcal{V}}{dt} = \delta \varphi \frac{Q^2}{\mathcal{I}} - c \mathcal{V}. \quad (65)$$

With a constant clock $\lambda = \delta$ instead, the mean export is $\delta \varphi Q$ and the bulk ODEs collapse to Case 3 with $\varepsilon_{\text{eff}} = \delta \varphi$: the pulses remain visible in single-cell trajectories and higher moments, but not in the first-moment ODE. Indeed, partial vs full release is invisible in mean ODEs whenever the event rate is constant; it becomes visible in bulk dynamics when the clock is load-dependent, and in single-cell release-count distributions always.

The recommendation: boolean thinning with a load-dependent clock ($\lambda = \delta X$ or δX^2) as the primary intermediate — one transparent parameter φ between mild partial clear and full dumping, mechanistic, nested in Cases 3–4 at the mean-field level, easy to simulate and to write BMVR ODEs for; fixed fraction as a secondary when less burst-size noise is wanted. The flooding boundary as a function of φ — where on this continuum the advantage of §5 survives — is open.

7 Contrast with HIV: what not to claim

The core results of this chapter are written for pathogens whose intracellular life ends in load-proportional catastrophe — lyse and dump. The motivating applications (*Y. pestis*, *F. tularensis*, lytic phage) are of that kind. HIV is not: it buds from the plasma membrane, and an infected cell need not rupture and empty itself in one event. Yet single-cell and limiting-dilution data show something that looks surge-like — long quiet periods, then a concentrated bout of release

— which is exactly the phenomenon the BDC explains. The honest position is neither to force HIV into the pure BDC nor to retreat to constant- p BMVR, but to build an eclipse-structured, short-productive-pulse model on the same renewal skeleton. This section records what the best data support, gives candidate equations, and draws the boundary of the chapter’s claims.

7.1 What Hataye et al. (2019) show

Hataye and co-authors [6] coupled limiting-dilution latency-reactivation cultures to stochastic models, and their findings — summarised here at the level that bears on equation choice — fall in two groups.

On initial release (viral inhibition cultures; no de novo infection).

1. The timing and magnitude of first HIV RNA release from reactivated latently infected CD4 T cells are highly variable: total yield, delay to first detection, and detection duration all fluctuate.
2. Mean release is delayed: first detection most often around day 3, with a sudden bulk of production around days 3–5 — not a flat continuous drip from day 0.
3. A simple productive-cell model (release at constant rate p , death at rate δ_I , total yield geometric with mean p/δ_I) is under-dispersed relative to the data: the Fano factor is too small.
4. A single exponential eclipse $E \rightarrow I$ fits the delay poorly (its mode sits at day 1). A multi-stage eclipse — an Erlang-like chain of n compartments, with $n \approx 5$ favoured — fixes the mean timing.
5. Multi-stage eclipse alone still under-disperses, and transcriptional ON/OFF toggling does not fix the dispersion under realistic CD4 division/death rates ($\rho_{\text{div}} \approx \mu_E \approx 0.5/\text{day}$ from CFSE and population stability data).
6. Their best structure (“Model 3”) uses two distinct eclipse paths (high- and low-release potential), multi-stage delay, and division/death in eclipse. Then: many stimulated lineages release nothing detectable ($f_{\text{HIV}} \approx 0.41$ of single latently infected cells give detectable virus in their fit); most productive sojourns last on the order of a day; and multi-day “sustained” detection is largely proliferation in eclipse followed by staggered short $E \rightarrow I$ transitions of descendants — not one cell budding steadily for a week.

On establishment (outgrowth cultures; de novo infection allowed).

1. Release of replication-competent virus often fails to establish exponential spread.
2. Establishment probability as a function of the expected number of virus-releasing cells per well is sigmoid, not the concave curve of independent per-cell success — interpreted as an Allee effect, synergy at low census.
3. A critical threshold sits near $\Lambda \approx 2.3$ latently infected cells, corresponding on average to roughly 5100 detected HIV RNA copies of initial release (under their establishment definition of $\geq 2 \times 10^5$ copies).

4. For exactly one stimulated latently infected cell, their fitted dynamics predict establishment only on the order of 2% of the time — consistent with the known gap between intact provirus counts and quantitative viral outgrowth.

Supported for HIV	Not supported as literal HIV
Multi-stage eclipse (delay)	Pure one-type BDC terminal catastrophe
Short productive pulse then cell death	Memoryless constant- p BMVR from $t = 0$
High stochasticity; many silent lineages	Independent branching alone near threshold
Renewal / stage structure for free virus	QSD = geometric burst size as an HIV law
Allee / synergistic establishment	

The visual “surge” is real, but in their best model it is delay plus brief constant-rate production plus latent lineage dynamics — not a single lytic dump of intracellular load.

	Lytic BDC (this chapter)	HIV (Hataye-guided)
Release mechanism	catastrophe at rate δX	budding while productive
End of production	terminal; process absorbed	short productive life; cell dies
Intracellular load X	explicit birth–death state	not required as primary state
Quiet period	until first catastrophe	multi-stage eclipse (+ latency)
Sustained multi-day release	one large dump (or nothing)	often staggered short pulses
Establishment theory	independent offspring as baseline	Allee synergy near threshold
Natural hosts	<i>Y. pestis</i> , <i>F. tularensis</i> , phage	CD4 T-cell latency / rebound

7.2 Candidate HIV stage equations

A compact Hataye-aligned system, suitable for coupling to the renewal free-virus layer of Result 3.1 (Figure 23). Let L_0 be a stimulated latently infected cell (or the start of reactivation). Two eclipse pathways run in parallel: A , of high release potential and multi-stage, and B , low and brief:

$$L_0 \xrightarrow{\alpha_A} E_1^{(A)}, \quad L_0 \xrightarrow{\alpha_B} E^{(B)}, \quad (66)$$

$$E_j^{(A)} \xrightarrow{\alpha} E_{j+1}^{(A)} \quad (j = 1, \dots, n-1), \quad E_n^{(A)} \xrightarrow{\alpha} I^{(A)}, \quad E^{(B)} \xrightarrow{\alpha_B} I^{(B)}. \quad (67)$$

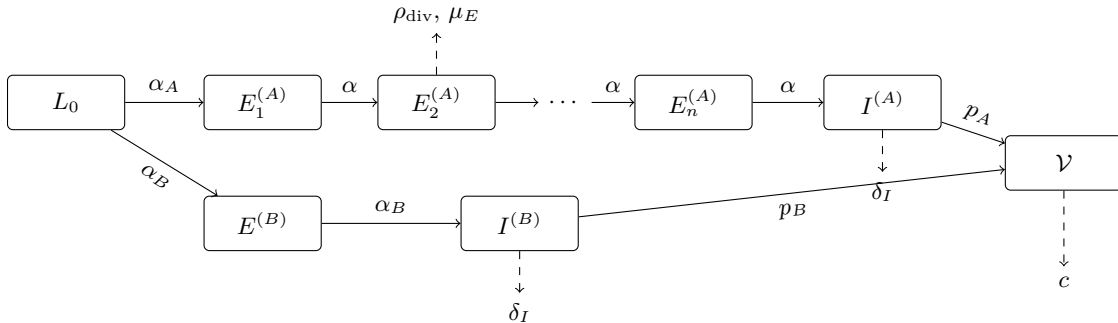


Figure 23: The Hataye-aligned stage structure of §7.2: two eclipse pathways from the stimulated latent class L_0 — a multi-stage path A of high release potential and a brief path B — each feeding a short productive class that releases at constant rate; division and death (dashed) act in eclipse, death δ_I in the productive classes, and clearance c on free virus.

In each eclipse compartment, cells may divide or die without releasing virus,

$$E \xrightarrow{\rho_{\text{div}}} E + E, \quad E \xrightarrow{\mu_E} \emptyset, \quad (68)$$

with $\rho_{\text{div}} \approx \mu_E \approx 0.5/\text{day}$ in the Hataye fits. Productive cells release at constant rate and die,

$$I^{(A)} \xrightarrow{p_A} I^{(A)} + \mathcal{V}, \quad I^{(A)} \xrightarrow{\delta_I} \emptyset, \quad I^{(B)} \xrightarrow{p_B} I^{(B)} + \mathcal{V}, \quad I^{(B)} \xrightarrow{\delta_I} \emptyset, \quad (69)$$

with $p_A \gg p_B$ (or p_B small and $I^{(B)}$ short-lived) to match the bimodal release distribution; free virus clears at rate c . The mean-field stage ODEs (large inoculum Λ , suppressing the dual path for display) are

$$\frac{dE_1}{dt} = \alpha_A L_0 \delta_0(t) + \rho_{\text{div}} E_1 - (\mu_E + \rho_{\text{div}} + \alpha) E_1, \quad (70)$$

$$\frac{dE_j}{dt} = \alpha E_{j-1} + \rho_{\text{div}} E_j - (\mu_E + \rho_{\text{div}} + \alpha) E_j, \quad j = 2, \dots, n, \quad (71)$$

$$\frac{dI^{(A)}}{dt} = \alpha E_n - \delta_I I^{(A)}, \quad \frac{d\mathcal{V}}{dt} = p_A I^{(A)} - c \mathcal{V} \quad (+ p_B I^{(B)}), \quad (72)$$

where the Dirac source stands for an initial cohort; for ongoing infection it is replaced by incidence, split between pathways with $\pi_A + \pi_B = 1$:

$$i(t) = f(\mathcal{V}(t), T), \quad \text{source into } E_1^{(A)} = \pi_A i(t), \quad \text{source into } E^{(B)} = \pi_B i(t). \quad (73)$$

Linear infection (the classical, large-population limit) takes $f(\mathcal{V}, T) = \gamma T \mathcal{V}$ and recovers a stage-structured BMVR. Near the establishment threshold, however, the Hataye data favour a sigmoid law over independent per-virion success; a minimal phenomenological upgrade is the Allee incidence

$$f(\mathcal{V}, T) = \gamma T \mathcal{V} \cdot \frac{\mathcal{V}}{K_A + \mathcal{V}}, \quad (74)$$

inefficient at very low $\mathcal{V}$ and saturating to mass action for $\mathcal{V} \gg K_A$, with K_A chosen so that cumulative initial release near $\sim 5 \times 10^3$ RNA copies sits at the inflection of establishment probability — the empirical threshold above. More mechanistic forms (cooperative entry, restriction-factor saturation, local multiplicity) can replace it; the point is that linearity in $\mathcal{V}$ alone is not faithful near establishment.

Coupled to free virus, the stage model induces renewal kernels of the usual kind. Let $g_{\text{HIV}}(a)$ be the expected release rate at age a after a single infection or reactivation, and $S_{\text{HIV}}(a)$ the probability of still being in a productive (or not-yet-cleared) lineage. Then exactly the skeleton (48) holds:

$$\mathcal{I}(t) = \mathcal{I}_0 S_{\text{HIV}}(t) + (i * S_{\text{HIV}})(t), \quad \frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g_{\text{HIV}}(t) + (i * g_{\text{HIV}})(t) - c \mathcal{V}(t). \quad (75)$$

For the stage model, g_{HIV} is *not* δK : it is the density of productive sojourn times times p_A (and p_B), with the Erlang eclipse supplying the delay — $g_{\text{HIV}}(a) \approx 0$ for small a , then rising and falling: the mathematical expression of “nothing for a long time, then a lot”. The effective parameters of §4 carry over by the substitution $(I_{\text{fix}}, g) \mapsto (S_{\text{HIV}}, g_{\text{HIV}})$:

$$p_{\text{eff}}(r) = \frac{\tilde{g}_{\text{HIV}}(r)}{\tilde{S}_{\text{HIV}}(r)}, \quad R_0 = \frac{\gamma T}{c} \int_0^\infty g_{\text{HIV}}(a) da \quad (\text{linear incidence}), \quad (76)$$

while under the Allee form R_0 remains the large-population threshold and early establishment is governed in addition by the scale K_A .

7.3 What to keep, what not to claim

- **Keep** for HIV: the renewal free-virus coupling; age/stage kernels; the lesson that constant p is only an exponential-phase projection; stochastic early failure; eclipse as a real dynamical object (the GATE mechanism of Chapter 5, multi-stage here).
- **Do not claim** for HIV: terminal catastrophe at rate δX ; QSD = geometric burst size as an HIV law; one-shot cumulative release as the only single-cell trajectory; independent branching as the full establishment story near the Hataye threshold.
- **Flooding comparison:** still meaningful as a *linear* baseline — matched mean yield, compare offspring variance — but near $\sim 5 \times 10^3$ RNA copies synergy can dominate variance effects, and any HIV write-up should separate the two mechanisms.

7.4 HIV intracellular mean growth: linear vs exponential

One modelling decision underlies the stock-and-export cases of §6.2 and deserves its own statement: does the mean intracellular load grow exponentially (birth–death) or linearly (template-limited production)?

Regime	Mean law	Mechanism
Mean-exponential	$m' = r m$	each unit generates more units (branching / BDC)
Mean-linear	$m' \approx \nu - \mu m$	production set by fixed templates / machinery

Absorbing BDC and pure birth–death are exponential-mean; immigration and constant production are linear-mean. HIV does *not* amplify infectious units inside the cell by branching: after integration the DNA template count is typically one or a few proviruses, not an exponentially growing genome pool; late genomic RNA is made by host Pol II from that fixed template, not by cytoplasmic genome cascading; and Gag and the other structural products are translated from mRNA and do not self-replicate. Production is immigration-like: rate set by mRNA and ribosomes, minus degradation, packaging and export. Pure $m' = r m$ is a poor literal description of HIV intracellular load; it fits phage genome races, intracellular bacterial division, and some cytoplasmic RNA viruses far better. The main nonlinearity is Tat positive feedback: transcription rises from a low basal level towards a saturated high state. During the switch, product counts can rise faster than a pure linear ramp, because the *production rate* is increasing — not because each unit is a branching particle; and once fully ON, production is roughly constant about a high mean, so mean accumulation is linear or saturating. The late fully-ON rate is the ceiling, and the intermediate phase climbs towards it; in the mean it does not systematically exceed it (stochastic transcriptional bursts can spike above the late *mean* rate, but that is noise, not a hotter phase; and net accumulation can peak mid-course if export turns on late, but that is export balance, not Tat overshoot). Confidence: high that HIV is not pure exponential-mean birth–death of virion-like units; medium–high that the main productive window is closer to mean-linear than exponential; medium on the exact shape of $m(a)$, which depends on the molecular species counted and on the Tat phase. The recommended coarse regime is therefore

$$\text{eclipse} \longrightarrow \text{mean-linear (immigration-type) accumulation of } X, \quad (77)$$

with a fixed or multi-step delay so first activity is not peaked at $t = 0^+$, production $m' \approx \nu - \mu m$ (or $m' \approx \nu$ while accumulating) at the late fully-ON rate ν , Tat switch-on folded into a slightly longer eclipse plus an abrupt start of ν , and the export law left open — the linear-vs-exponential

choice for accumulation does not force it. In ranking: birth–death / absorbing BDC fits poorly as a literal HIV mechanism; immigration plus export or dump is a much better caricature; eclipse then constant production rate is the best simple default; Tat ON/OFF plus immigration the best simple mechanistic default.

7.5 Suggested role of this material

The core lytic results stand as the chapter’s contribution; this section is their honest boundary. In a companion paper the natural order is: the BDC and renewal results at the lytic end of the spectrum; then “HIV is not BDC”, with Hataye’s Model 3 as the empirical intermediate and the stage equations above; then numerical extraction of $g_{\text{HIV}}(a)$ from the stage model (or from daily release curves), overlaid against constant- p BMVR exactly as in §3.4; and finally establishment by independent branching vs the Allee form, with the critical release scale taken from experiment.

8 Discussion

This chapter took the completed single-cell theory of Chapters 3 and 4a and embedded it in a population. The outcome is a renewal model of within-host infection forced by the intracellular process, exact in its kernels; the demonstration that the classical constant-rate model is a projection of it, valid only one exponential phase at a time; an effective-parameter map whose ends differ by orders of magnitude; the invariance of R_0 under the replacement of budding by bursting; the flooding advantage of bursting at establishment, and its price in invasion speed; a spectrum of release models connecting the two ends; and the honest boundary of all of it, drawn against HIV. What remains is to say what the results imply for the way models are fitted, how they connect to the chapters that follow, and what is still open.

8.1 Consequences for fitting practice

Three consequences for the way within-host models are fitted to data. First, p is not a constant of the pathogen: the effective release rate $p_{\text{eff}}(r)$ falls monotonically with the epidemic growth rate, from $V_{\infty}/\mathbb{E}(T_{\text{prod}})$ in the stationary phase towards δ in the acute phase, and fits to acute and set-point data should therefore yield systematically different estimates, in a direction and by an amount the theory specifies (§4). Second, the eclipse-like delay usually inserted into within-host models by hand emerges from the intracellular dynamics with no extra parameter; its duration is the mean productive lifetime’s shadow, $\lambda^{-1} \log \langle X \rangle_{\text{QS}}$ ((13)). Third, the identifiability negative result of §4.3 says that no fit of (p, d_I) — from any data, at any growth rate — can recover the three intracellular rates; a third observable is required, and the theory names the three cheapest ones: the burst-size dispersion (geometric ratio $1/a$), the no-burst fraction b , and the shape of the release-time density. An estimate of p alone tells one almost nothing about the cell; an estimate of p plus any one of the three tells one something testable. The single-cell assays that measure those three observables directly are described in the discussion of Chapter 4a, and the biological systems where the estimates have already been pushed to both extremes — *F. tularensis* and *B. anthracis* — are weighed there as well.

8.2 Connections forward

The two-type process of Chapter 5 supplies the most immediate continuation. In the GATE regime $\delta_1 = 0 < \delta_2$ — no burst before conversion — the no-catastrophe probability has a zero initial hazard, then rising: that is precisely the eclipse-phase signature which within-host models insert by hand, emerging mechanistically, with the eclipse duration set by the conversion rate ν . The renewal construction of this chapter is ready to receive the two-type kernels; what blocks the way is the two-type calculation itself (below). The budding–bursting comparison of §5 finds its evolutionary home in Chapter 6, where the spectrum of §6 — partial release, reset catastrophe, self-excitation — provides the intermediate phenotypes a selection argument needs; and the nonlinear mechanisms Chapter 7 discusses in the Komarova vein (immune saturation, local depletion, the “all run out at once” intuition) are exactly the effects the branching-structure comparison of §5 deliberately excludes, and should be named wherever the flooding advantage is invoked.

8.3 Open problems

Five questions remain open on the population side.

1. **Two-type killed second moments and the two-type D .** The release kernel of a two-type renewal model needs $\mathbb{E}(X_a^2 \mathbf{1}\{\tau_c > a\})$, $\mathbb{E}(X_a Y_a \mathbf{1}\{\tau_c > a\})$, $\mathbb{E}(Y_a^2 \mathbf{1}\{\tau_c > a\})$, and the analogue of $D(t)$ (extinction without catastrophe). The moment hierarchy does not close upward, exactly as in one type; the one-type escape was to differentiate the Riccati equation, and the two-type route should be to differentiate the backward system of Chapter 5. This is the main technical obstacle between this chapter and an eclipse-aware renewal BMVR, and it should not be assumed easy.
2. **The flooding boundary against real parameters.** Map $\delta(\lambda + \mu) > \lambda\mu$ for *Y. pestis* in the macrophage, and compute the *size* of the advantage from (43): does the pathogen sit on the advantageous side, and by how much? The estimates discussed in Chapter 4a — the near-certain bursting of *F. tularensis*, the clearance-dominated anthrax case — suggest that real pathogens sit on both sides of the boundary, which is what makes the question worth asking.
3. **Population-level variance.** Propagate $\text{Var}(W)$ of (12) through the branching structure to a whole-host prediction: for single-cell assays the first moments suffice, but for whole-host data the variance is the main stochastic signal the BMVR cannot reproduce. The integer-valued nature of the offspring count, whose distribution matters for establishment [7], enters through the same quantity.
4. **Partial release and the flooding boundary in φ .** Where on the boolean continuum of §6.5 does the flooding advantage survive? Does the closed-form structure persist at intermediate φ ?
5. **Literature positioning.** Verify whether Carruthers et al. [8] already couple an intracellular BDC to a between-cell model — the novelty claim of §3 depends on the answer — and position the renewal embedding against the Pearson–Krapivsky–Perelson [9], Nelson–Gilchrist–Perelson, Gilchrist–Coombs [10], and Heffernan–Wahl lines of work, with the BMVR’s attribution settled between McLean [1], Nowak–Bangham [2], and Perelson et al.

The single-cell open problems — a conceptual proof that burst size equals the quasi-stationary distribution, the chained joint laws for general μ , and the sensitivity of the kernels to logistic intracellular growth — are stated in the discussion of Chapter 4a, where they belong. The two lists together are the research agenda that this part of the thesis leaves in good order: each problem is named, and each has a route.

A Master formula table: population level

Throughout: $w = e^{\theta t}$, $\theta = \lambda(a - b)$, $A = a - 1$, $B = 1 - b$, $L = a(1 - b)$, $\kappa = 1 + \delta/(2\lambda)$. The single-cell rows are collected in Chapter 4a's Appendix A; this table records the population-level quantities of the present chapter.

Quantity	Formula
Renewal system	$\mathcal{I} = \mathcal{I}_0 I_{\text{fix}} + i * I_{\text{fix}}, \quad \mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}, \quad i = \gamma T \mathcal{V}$
Survival kernel	$S(a) = I_{\text{fix}}(a)$
Release kernel	$g(a) = \delta K(a) = V'(a)$
$p_{\text{eff}}(r)$	$\delta \tilde{K}(r) / \tilde{I}_{\text{fix}}(r)$
$d_{\mathcal{I}, \text{eff}}(r)$	$1 / \tilde{I}_{\text{fix}}(r) - r$
Limits $r = 0$	$p_{\text{eff}} = \lambda V_{\infty} / \log(a/(a - 1)), \quad d_{\text{eff}} = \lambda / \log(a/(a - 1))$
Limits $r \rightarrow \infty$	$p_{\text{eff}} \rightarrow \delta, \quad d_{\text{eff}} \rightarrow \mu + \delta$
Characteristic eq.	$r + c = \gamma T \delta \tilde{K}(r)$
R_0^{ODE}	$\gamma T V_{\infty} / c$
Branching mean	$m = q V_{\infty}, \quad q = \gamma T / (\gamma T + c)$
$G_{\text{off}}(z)$	$b + (\delta/\lambda) y / (a - y), \quad y = 1 - q + qz$
$z_{\text{ext}}^{\text{burst}}$	$[a(1 - q + qb) - 1 + q] / q$
$z_{\text{ext}}^{\text{bud}}$	$1/m$ (geometric output, matched mean)
Flooding criterion	$1/\delta < 1/\lambda + 1/\mu \quad (L > 1)$
$L = 1$	offspring law exactly geometric
Δz_{ext}	$(L - 1)(1/m - 1)$
$\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}}$	$2q^2 V_{\infty} (L - 1) / (a - 1)$
Growth trade-off	$r_{\text{bud}} > r_{\text{burst}}$ at matched $R_0, \mathbb{E}(T_{\text{prod}})$
HIV stage model	kernels $(S_{\text{HIV}}, g_{\text{HIV}})$; $p_{\text{eff}} = \tilde{g}_{\text{HIV}} / \tilde{S}_{\text{HIV}}$; Allee incidence $f = \gamma T \mathcal{V}^2 / (K_A + \mathcal{V})$

Quick-reference specialisations

$\mu = 0$. $b = 0$, $a = 1 + \delta/\lambda$, $D \equiv 0$, $I_{\text{fix}} = I$, $V_{\infty} = 1 + \lambda/\delta = \langle X \rangle_{\text{QS}}$, $\mathbb{E}(T_{\text{prod}}) = \lambda^{-1} \log(1 + \lambda/\delta)$, $\mathbb{P}\{\mathcal{K} = k\} = s \rho^{k-1}$ with $s = \delta/(\lambda + \delta)$, $\rho = 1 - s$; flooding favours bursting for establishment at every $\delta > 0$.

The boundary $L = 1$. $\delta = \lambda\mu/(\lambda + \mu)$, $b = \delta/\lambda$; the offspring law is geometric with mean m ; $z_{\text{ext}}^{\text{burst}} = z_{\text{ext}}^{\text{bud}}$ for every q ; offspring variances equal.

Young-cell / fast-growth limit. $p_{\text{eff}} \rightarrow \delta$, $d_{\text{eff}} \rightarrow \mu + \delta$: a newly infected cell holds one particle, bursts at rate δ , releases one.

Set-point / $r = 0$ matching. $p_{\text{eff}} = V_{\infty}/\mathbb{E}(T_{\text{prod}})$, $d_{\text{eff}} = 1/\mathbb{E}(T_{\text{prod}})$: lifetime output over lifetime; the fairest constant-parameter comparison of §3.4.

B Verification record: renewal BMVR

Every quantitative claim of this chapter was checked analytically where possible and numerically where not. This appendix records the numerical suite for the renewal system, its tally, and how to reproduce it.

The suite `verify_result_20_1.py` tests the renewal system (26)–(27) at four levels: the integrity of the kernels; the reduction to single-cell formulae without reinfection; the structural limits that can be checked in closed form; and an independent stochastic simulation of a finite cohort. Parameter sets: $(\lambda, \mu, \delta) \in \{(1, 0, 0.1), (1, 0.2, 0.05), (1, 0, 0.5), (1, 0.9, 0.1), (1, 0.5, 1/3)\}$. Fifty-four checks, all passing.

ID	Name	What is checked	Result
A	Kernel self-consistency	$V' = g$, $\int g = V_{\infty}$, $\int I_{\text{fix}} = \mathbb{E}(T_{\text{prod}})$, $I' = -\delta J$	5/5
B	$\gamma T = 0$ cohort	$\mathcal{I} = \mathcal{I}_0 I_{\text{fix}}$; $\mathcal{V}' = \mathcal{I}_0 g - c\mathcal{V}$ vs exact convolution; $c = 0 \Rightarrow \mathcal{V} = \mathcal{I}_0 V(t)$	15/15
C	V -equation residual	$\mathcal{V}' - \mathcal{I}_0 g - \gamma T(\mathcal{V} * g) + c\mathcal{V} \approx 0$ on supercritical / subcritical / virion-seed solutions	5/5
D	Exponential kernels $\equiv$ classical	$S = e^{-da}$, $g = pe^{-da} \Rightarrow$ renewal = BMVR ODE (rel. err. $< 5 \times 10^{-5}$)	3/3
E	Growth = characteristic root	late log $\mathcal{V}$ slope equals r_* solving $r+c = \gamma T \delta \tilde{K}(r)$ (rel. err. $\sim 10^{-6}$ – 10^{-7})	5/5
F	R_0 threshold	$R_0 \in \{0.5, \dots, 3\}$: decay iff $R_0 < 1$, growth iff $R_0 > 1$	8/8
G	$p_{\text{eff}}(r_*)$ match	classical ODE with $(p, d) = (p_{\text{eff}}(r_*), d_{\text{eff}}(r_*))$ shares r_* with the renewal system	4/4
H	Gillespie cohort	independent BDC cells + clearance; ensemble mean vs renewal (rel. $L^2 \sim 1$ –2%)	2/2
I	Virion-only seed	early $\mathcal{V} \approx V_0 e^{-ct}$; $\mathcal{I}(0) = 0$ then rises	1/1
J	Lifetime output	$\gamma T = c = 0$: $\mathcal{V}(\infty) = V_{\infty}$ (rel. err. $< 10^{-6}$)	5/5
K	Well-posedness	non-negative, finite trajectories	1/1
Total			54/54

One honest gap: a full stochastic multi-generation Gillespie check with reinfection ($\gamma T > 0$) is not yet coded. The reinfection case is already constrained by tests C, E, F, G through the characteristic equation, which any correct linear theory must satisfy, and agreement is expected by linearity.

Reproduction.

```
cd "4 BDC additional and BMVR/new_notes3"
python3 verify_result_20_1.py
# writes verify_result_20_1_report.txt, metrics, verify_figures/*.pdf
```


C Technical derivations

C.1 Hypergeometric Laplace transforms of the kernels

Write $s := r/\theta$ and recall $\theta = \lambda(a - b)$. Both transforms reduce, under $v = e^{-\theta t}$ (so $dt = -dv/(\theta v)$, $v : 1 \rightarrow 0$), to moments of the form $\int_0^1 v^{s-1}(1-v)^n(A+Bv)^{-m}dv$, which are hypergeometric. For the survival kernel, from (4),

$$I_{\text{fix}}(t) = \frac{(a-b)^2 w}{(B+Aw)(aw-b)}, \quad w = e^{\theta t},$$

and in $v = 1/w$,

$$\widetilde{I}_{\text{fix}}(r) = \frac{a-b}{\theta} \int_0^1 v^s \left[\frac{B}{A+Bv} + \frac{b}{a-bv} \right] dv,$$

using the partial-fraction decomposition $(a-b)/[(A+Bv)(a-bv)] = B/(A+Bv) + b/(a-bv)$ (both sides multiplied out give $(a-b)^2$ in the numerator after using $Ab + aB = a-b$). Each term is a standard hypergeometric integral, $\int_0^1 v^s/(1-zv)dv = (s+1)^{-1} {}_2F_1(1, s+1; s+2; z)$, giving

$$\widetilde{I}_{\text{fix}}(r) = \frac{1}{\lambda(s+1)} \left[\Psi(A, B, s) + \frac{b}{a} {}_2F_1\left(1, s+1; s+2; \frac{b}{a}\right) \right], \quad (78)$$

where, so that the hypergeometric series always converges,

$$\Psi(A, B, s) = \begin{cases} {}_2F_1(1, s+1; s+2; -\frac{A}{B}), & B \geq A, \\ \frac{B}{A} {}_2F_1(1, s+1; s+2; -\frac{B}{A}), & A > B, \end{cases} \quad (79)$$

the two branches being equal by the identity $(B/A) {}_2F_1(1, s+1; s+2; -B/A) = {}_2F_1(1, s+1; s+2; -A/B)$ — both sides equal $(s+1)B \int_0^1 v^s/(A+Bv)dv$. (The single-argument form $(B/A) {}_2F_1(1, s+1; s+2; -B/A)$, valid by analytic continuation for all $A, B > 0$, has a convergent defining series only when $B \leq A$, i.e. $\delta \geq \lambda - \mu$.) For the release kernel, $g = \delta K$ with $K = [1 + 2\lambda(1-I)/\delta]J$; the same substitution and termwise integration give

$$\delta \widetilde{K}(r) = \frac{\delta}{\lambda} \left[\frac{{}_2F_1(1, s; s+2; z)}{A(s+1)} + \frac{{}_2F_1(2, s; s+3; z)}{A^2(s+1)(s+2)} \right], \quad z = -\frac{B}{A}, \quad (80)$$

again with the understanding that for $B > A$ the defining series of the right-hand side must be evaluated through its analytic continuation (or through the integral representation from which it came). Both formulae reduce correctly at $s = 0$: with ${}_2F_1(1, 1; 2; z) = -\log(1-z)/z$, (78) gives $\widetilde{I}_{\text{fix}}(0) = \lambda^{-1} \log(a/(a-1)) = \mathbb{E}(T_{\text{prod}})$, and (80) gives $\delta \widetilde{K}(0) = (\delta/\lambda)(1/A + 1/A^2) = aB/A = V_\infty$, using $AB = \delta/\lambda$. Both reductions were verified numerically against direct quadrature, as was (80) for general r (relative error $\sim 10^{-8}$ in the convergent regime).

The companion derivations — the lifetime yield $V_\infty^{(k)}$ and the PGF coefficient extraction used by the chained model — are in Chapter 4a's Appendix C.

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